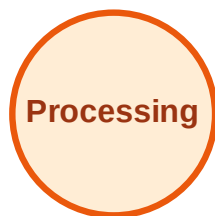


BFS Traversal

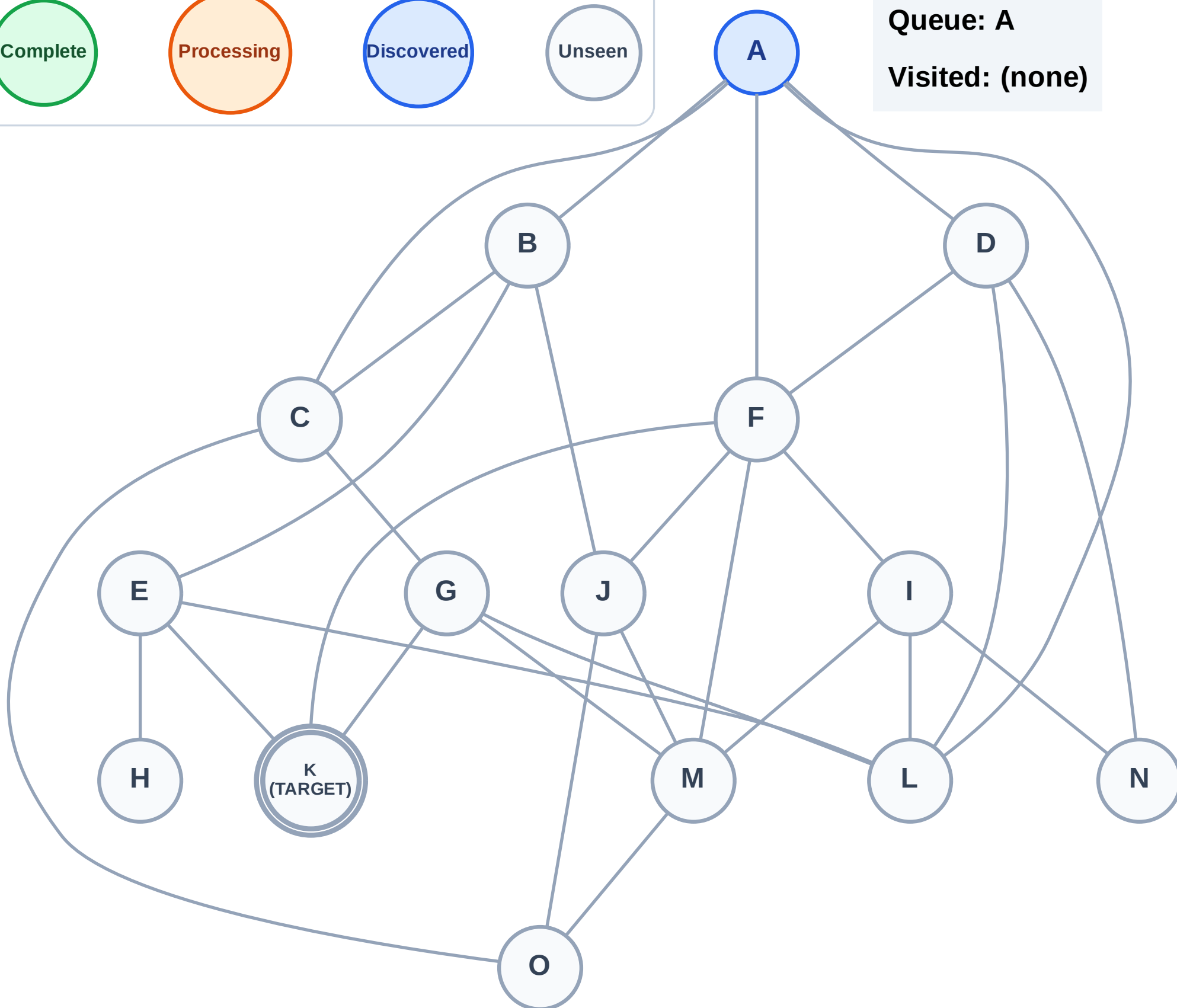
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

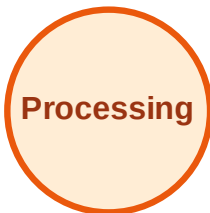
Visited: (none)



BFS Traversal

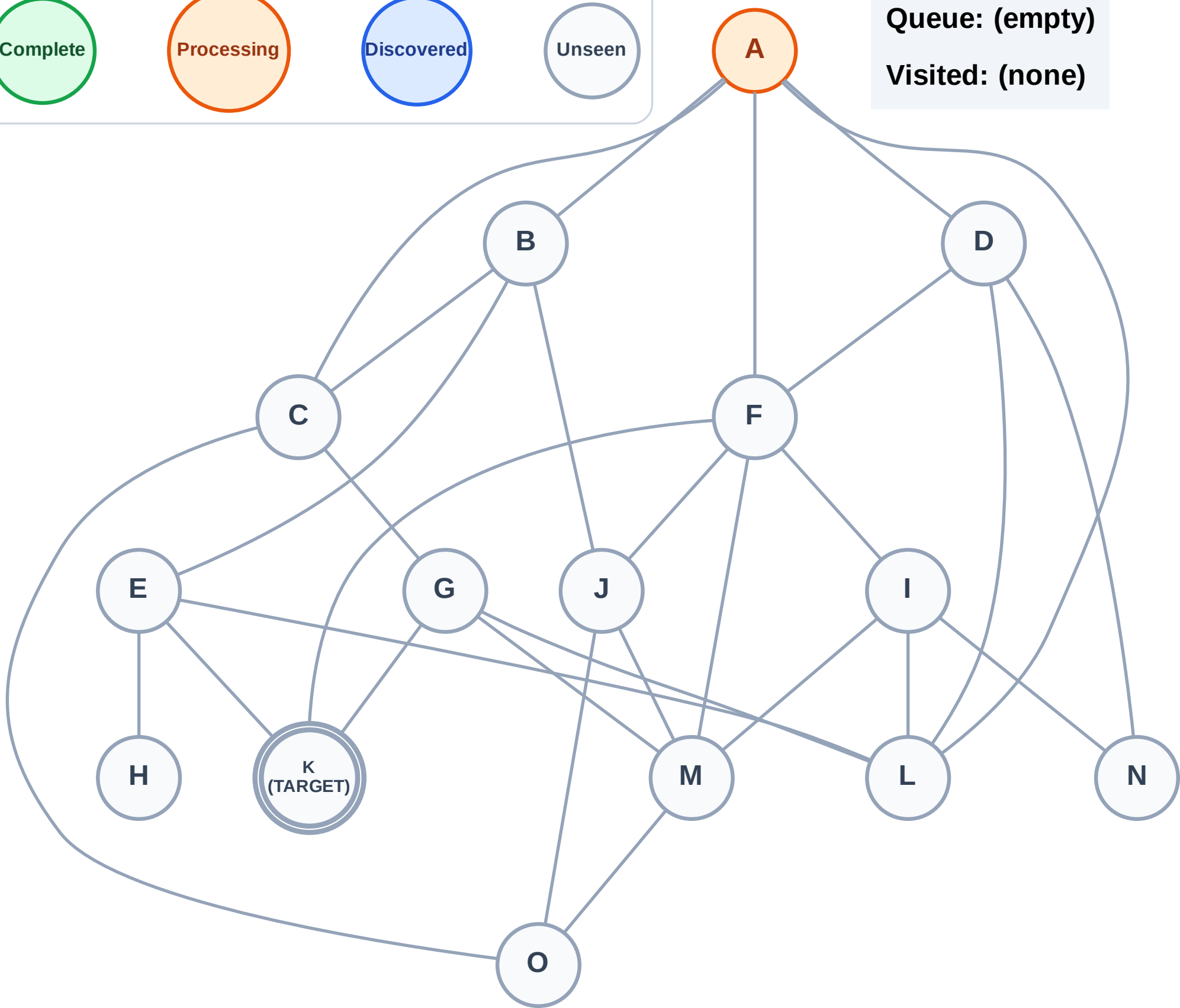
Step 2: Process node A

Legend



Queue: (empty)

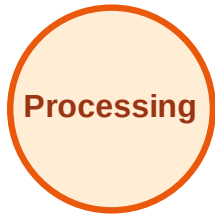
Visited: (none)



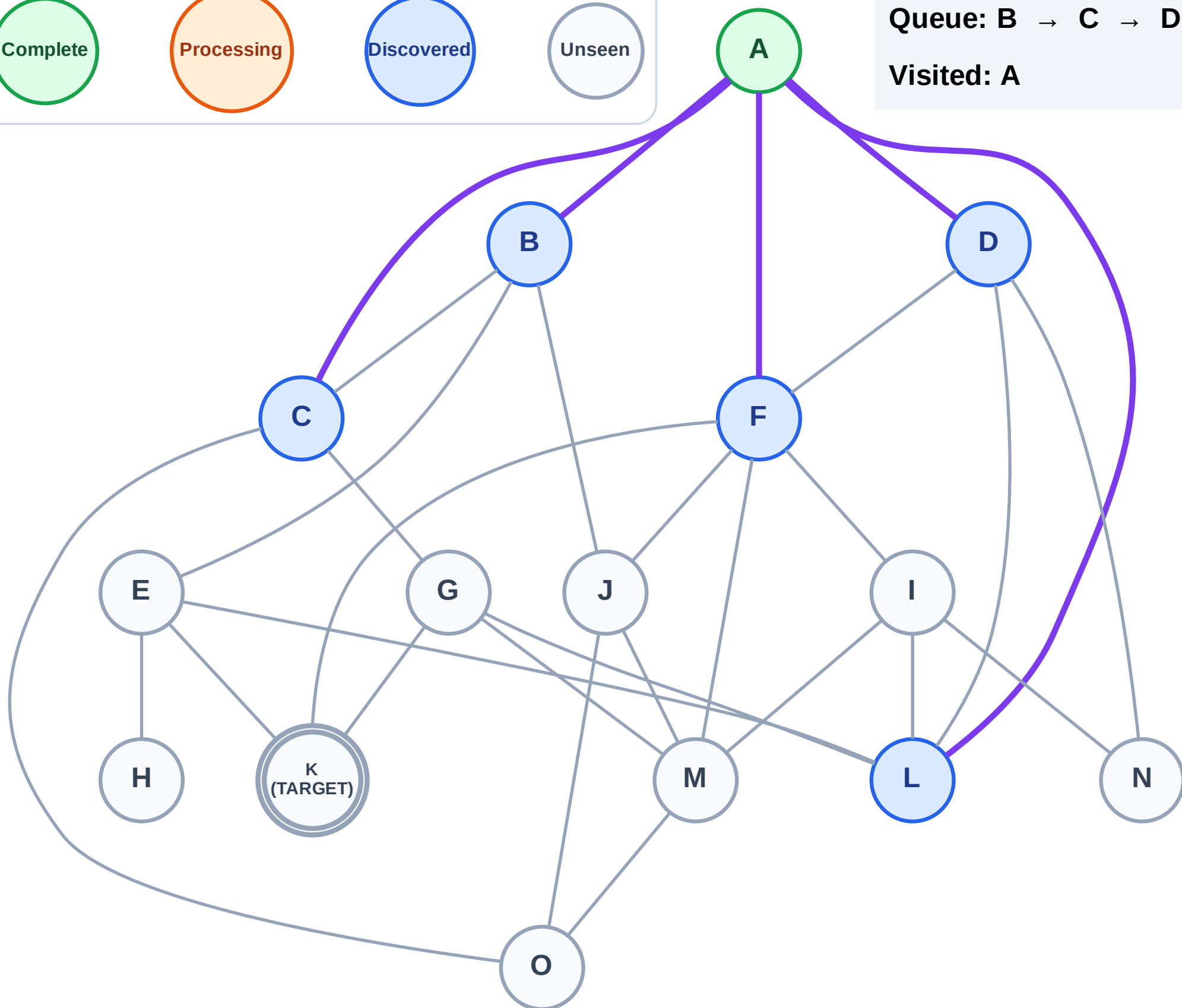
BFS Traversal

Step 3: Discover B, C, D, F, L from A

Legend



Queue: B → C → D → F → L
Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

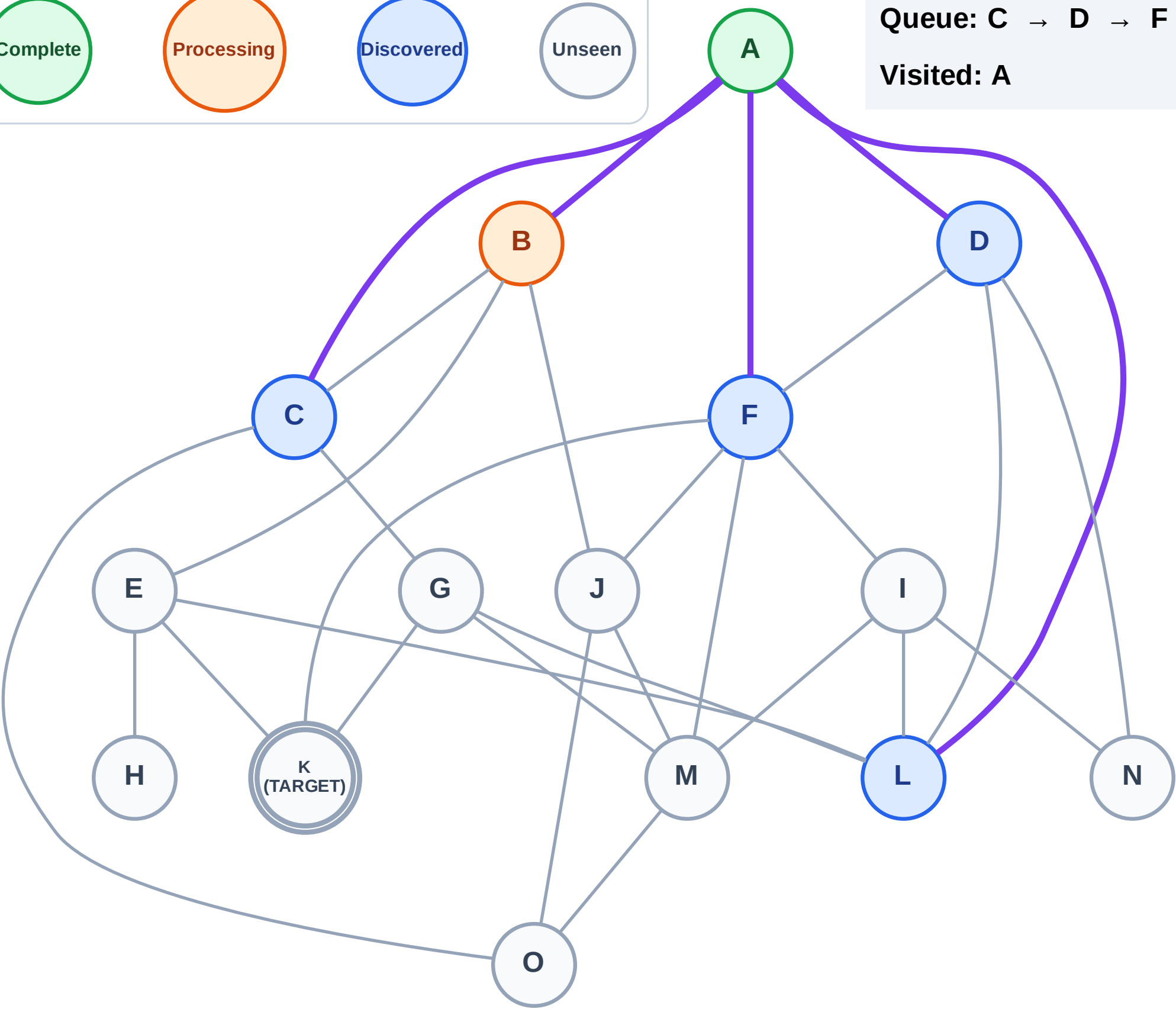
Processing

Discovered

Unseen

Queue: C → D → F → L

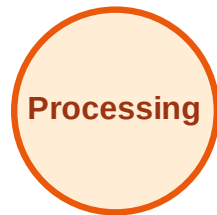
Visited: A



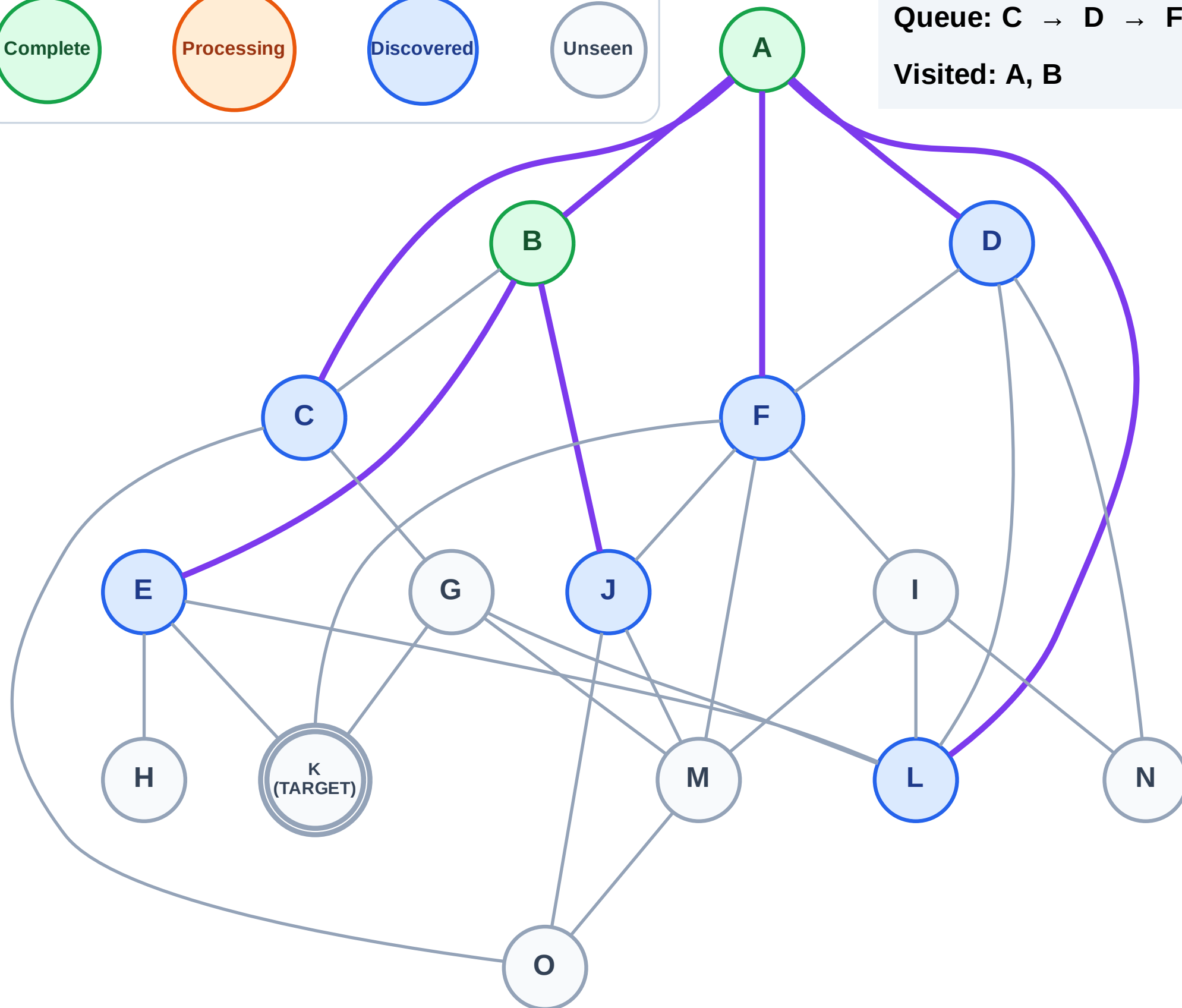
BFS Traversal

Step 5: Discover E, J from B

Legend

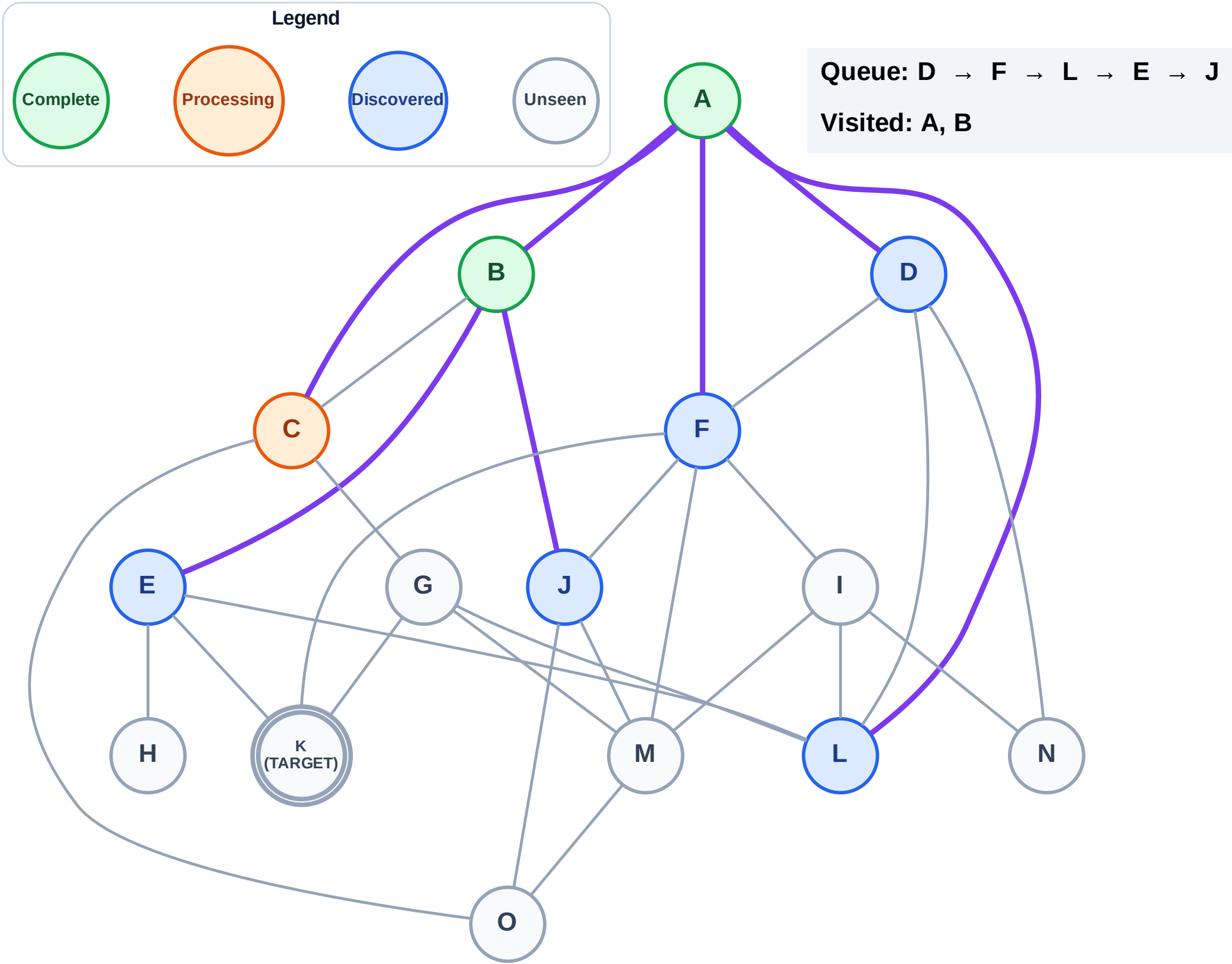


Queue: C → D → F → L → E → J
Visited: A, B



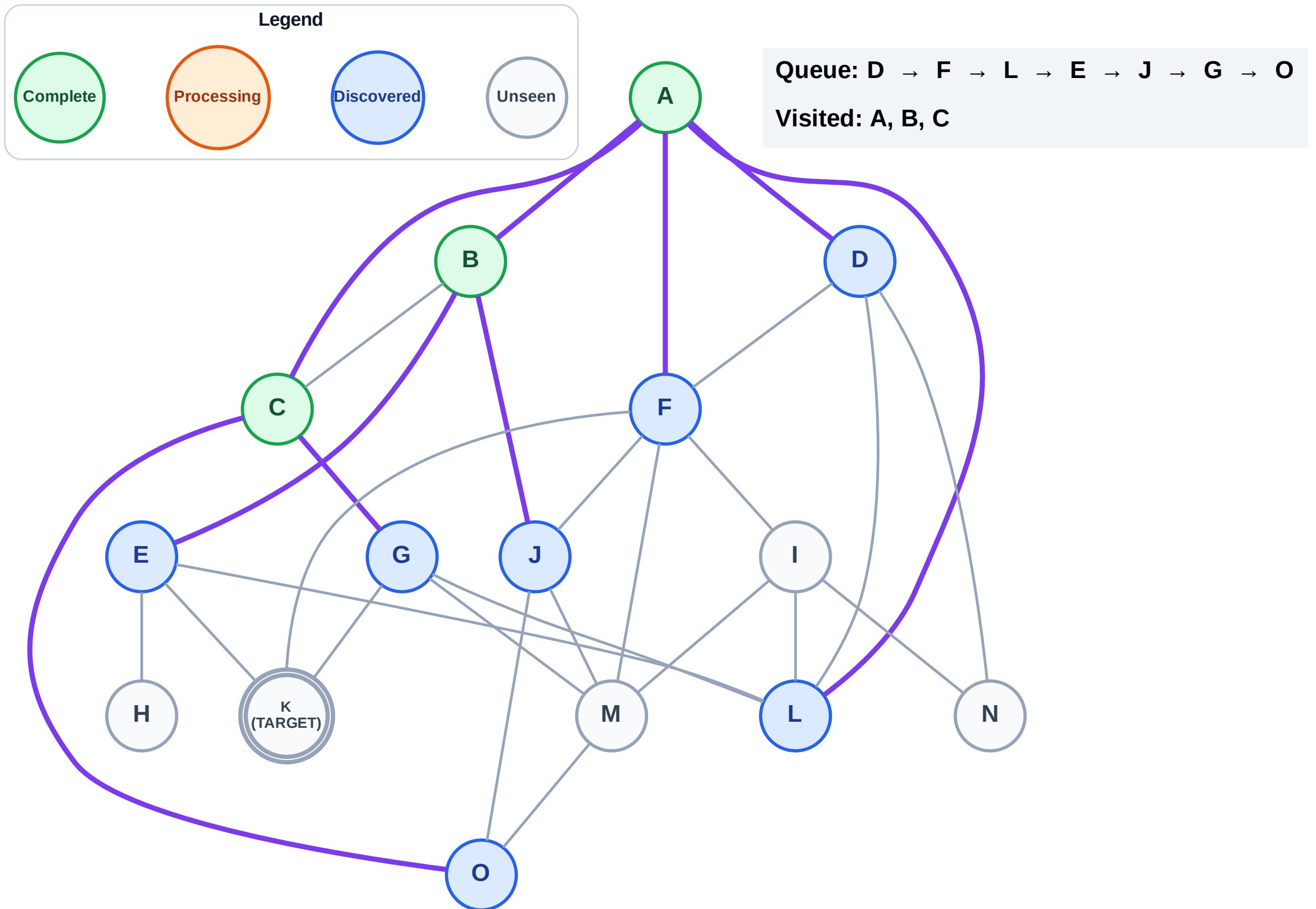
BFS Traversal

Step 6: Process node C



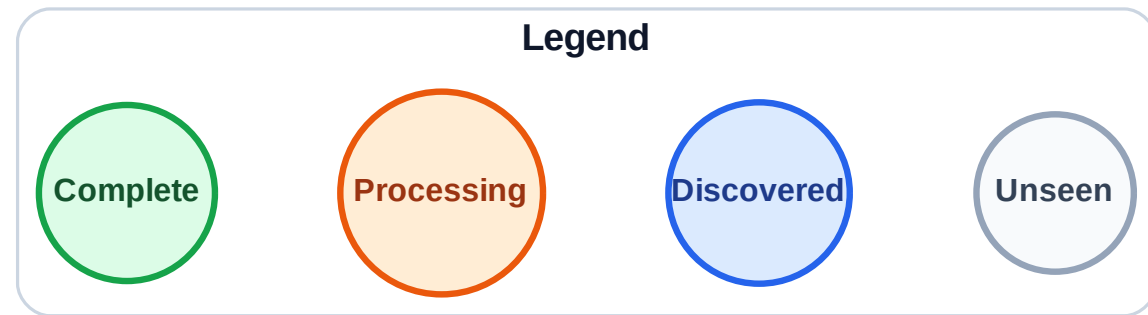
Step 7: Discover G, O from C

Step 7: Discover G, O from C



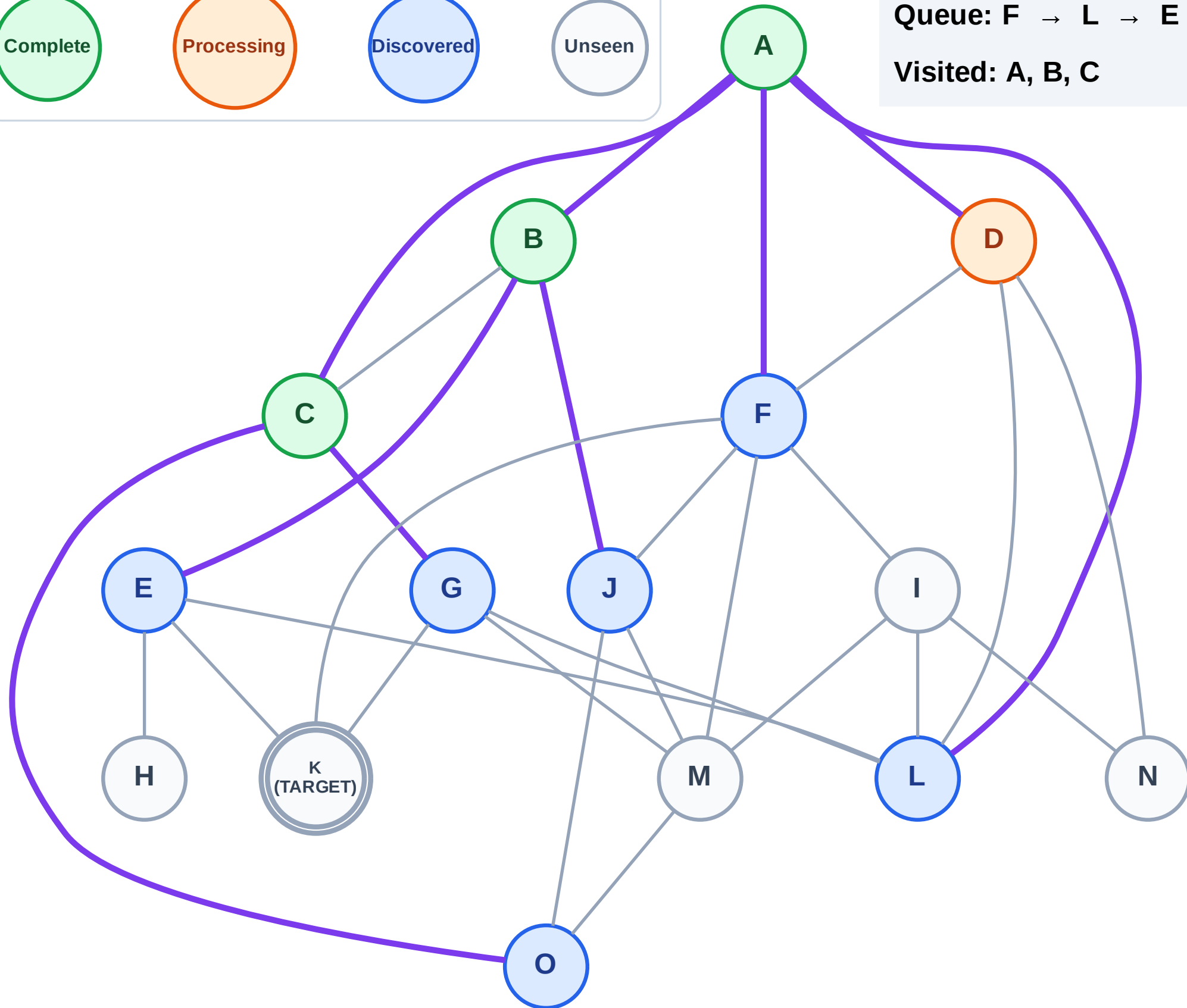
BFS Traversal

Step 8: Process node D



Queue: F → L → E → J → G → O

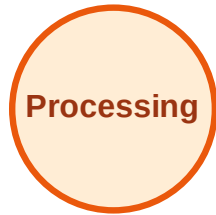
Visited: A, B, C



BFS Traversal

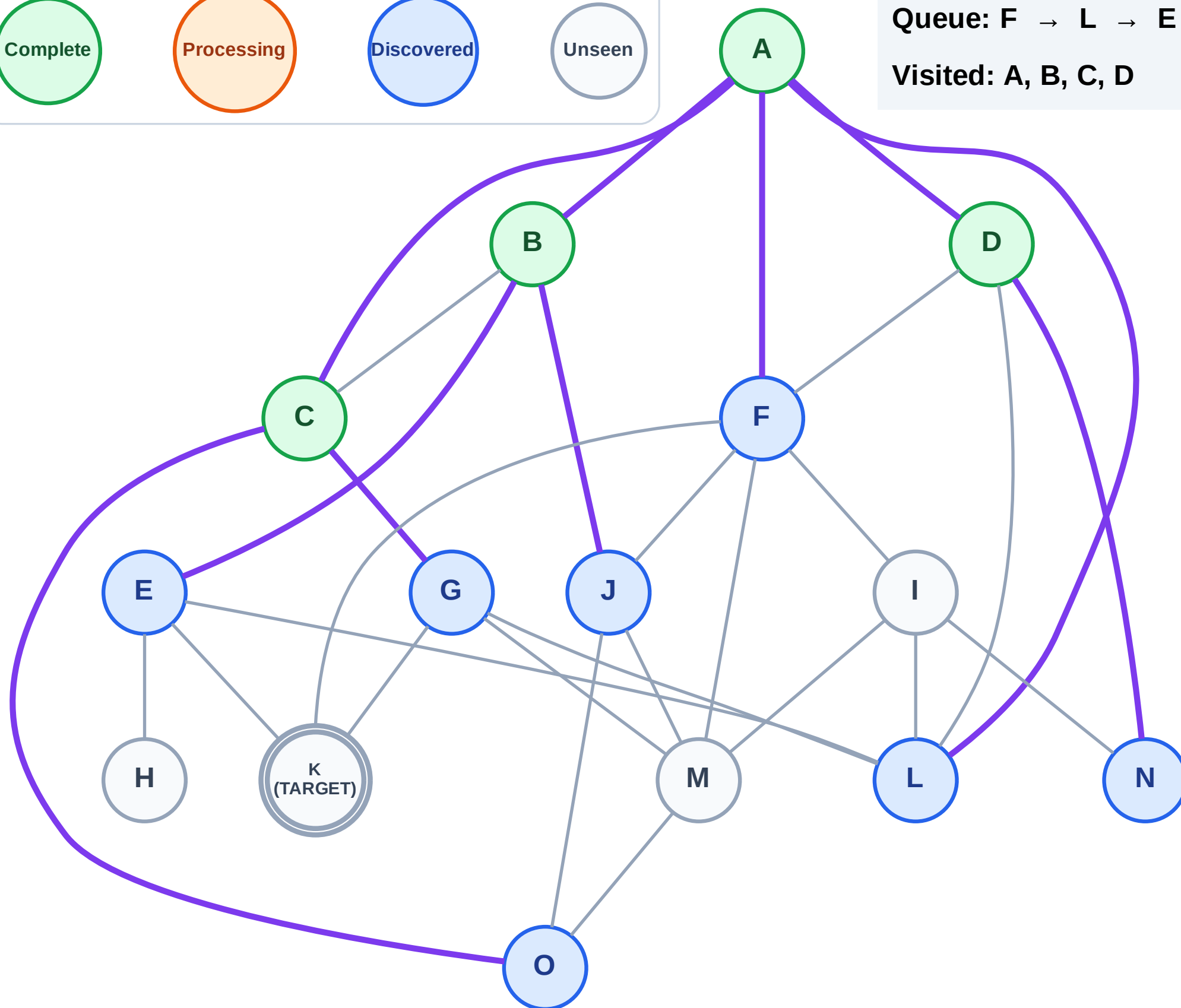
Step 9: Discover N from D

Legend



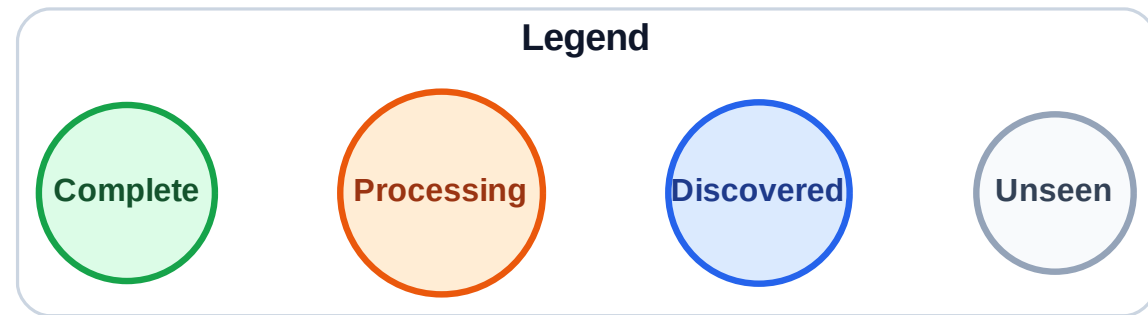
Queue: F → L → E → J → G → O → N

Visited: A, B, C, D



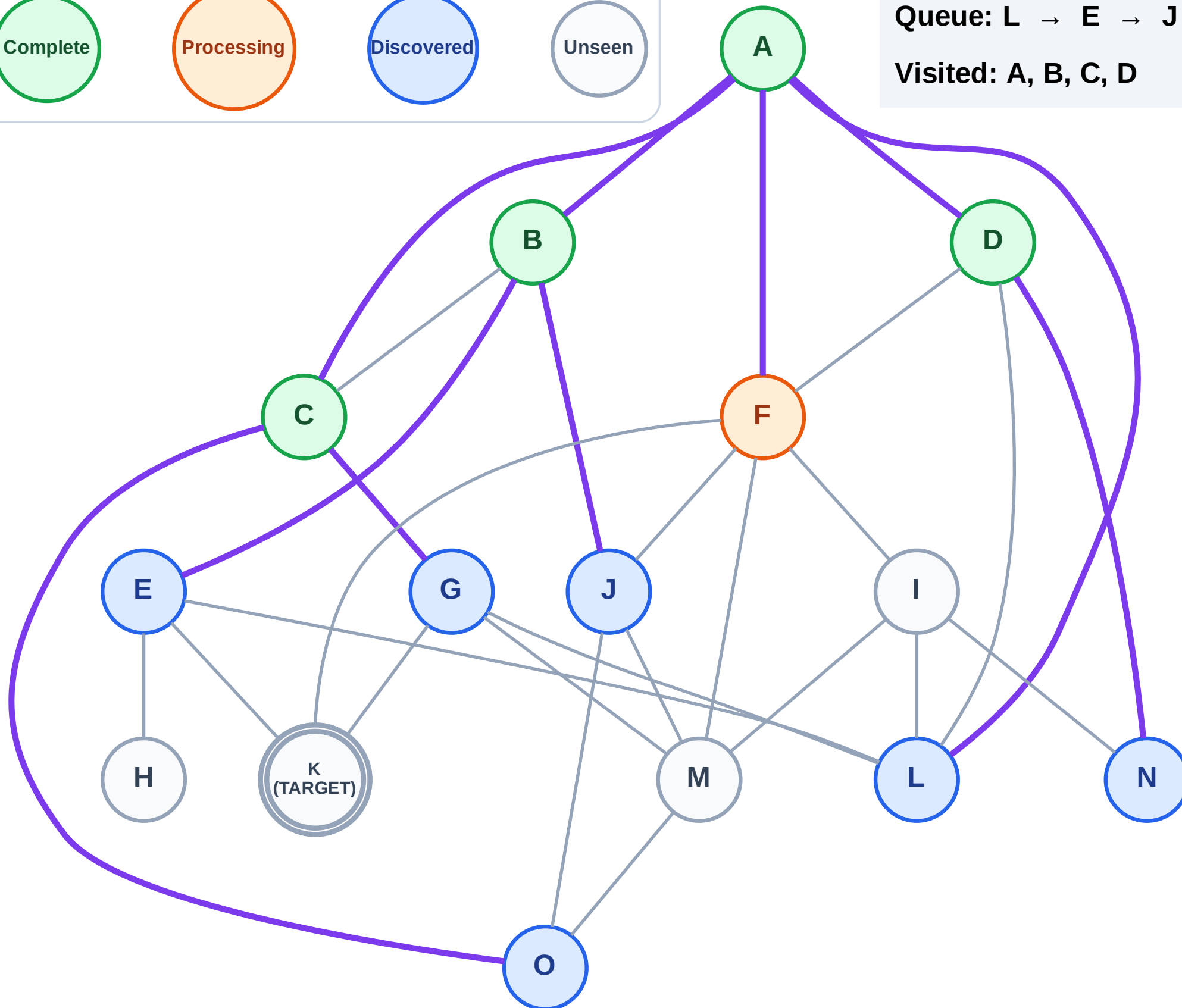
BFS Traversal

Step 10: Process node F



Queue: L → E → J → G → O → N

Visited: A, B, C, D



BFS Traversal

Step 11: Discover I, K, M from F

Legend

Complete

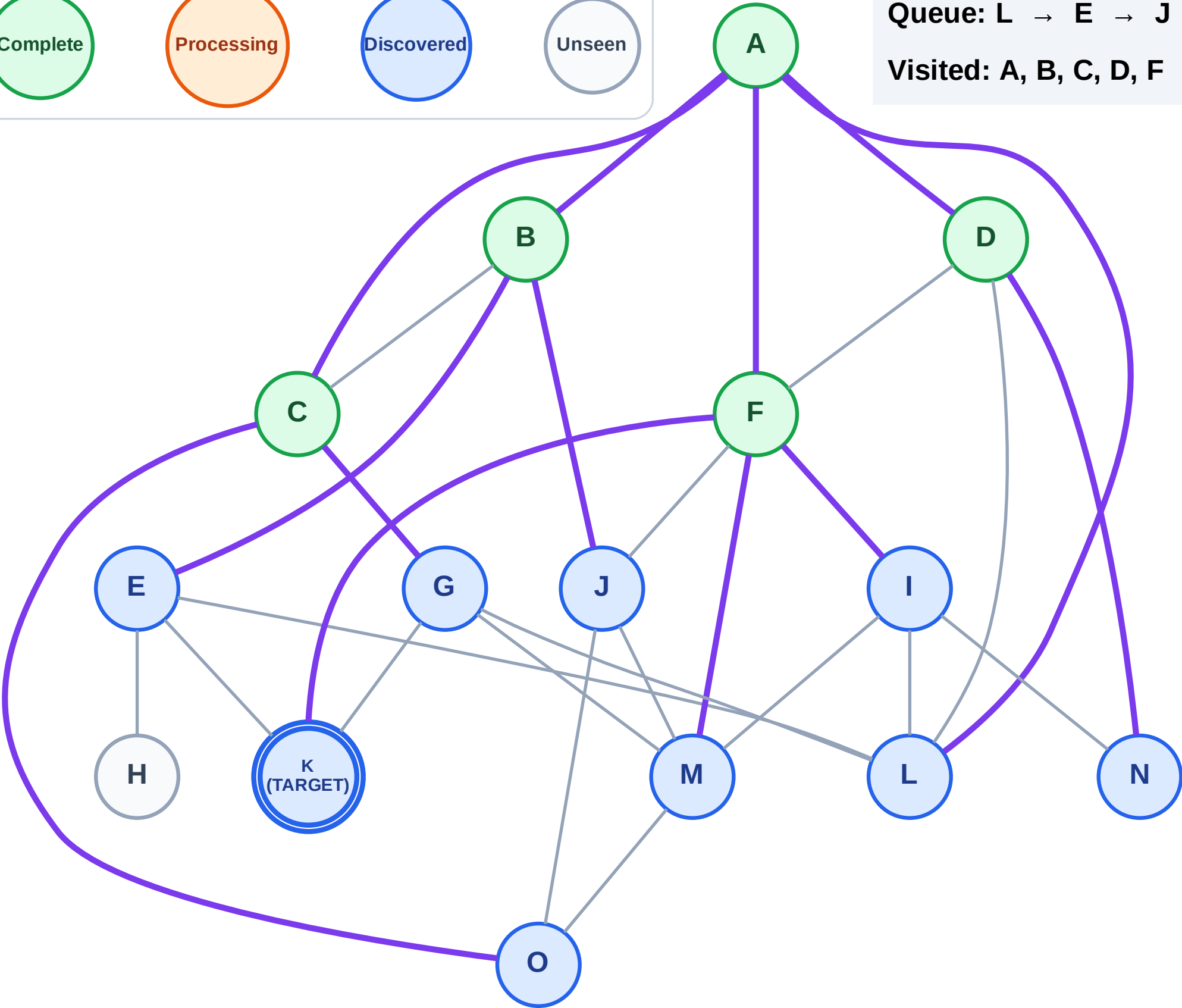
Processing

Discovered

Unseen

Queue: L → E → J → G → O → N → I → K → M

Visited: A, B, C, D, F



BFS Traversal

Step 12: Process node L

Legend

Complete

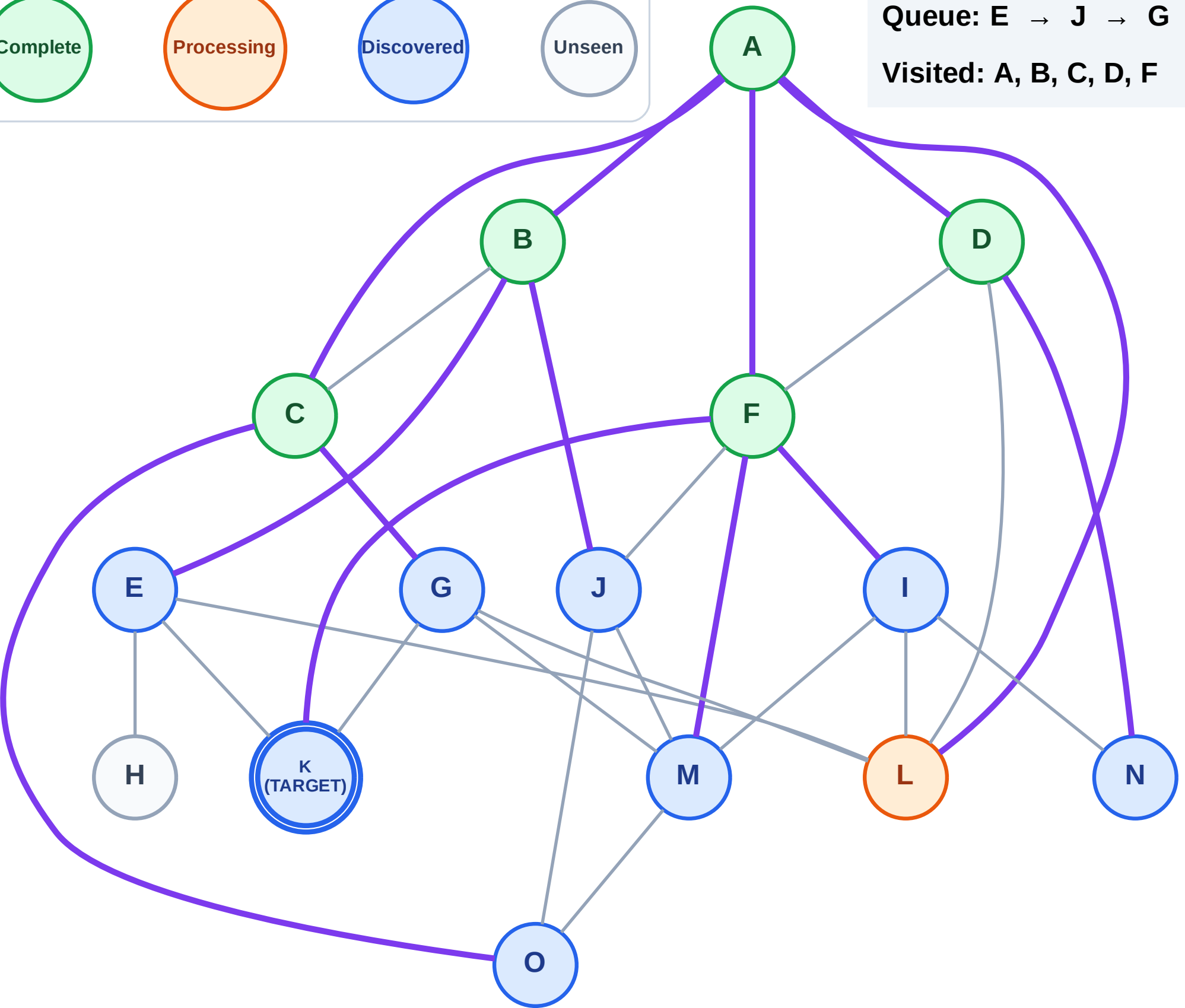
Processing

Discovered

Unseen

Queue: E → J → G → O → N → I → K → M

Visited: A, B, C, D, F



BFS Traversal

Step 13: No new neighbors from L

Legend

Complete

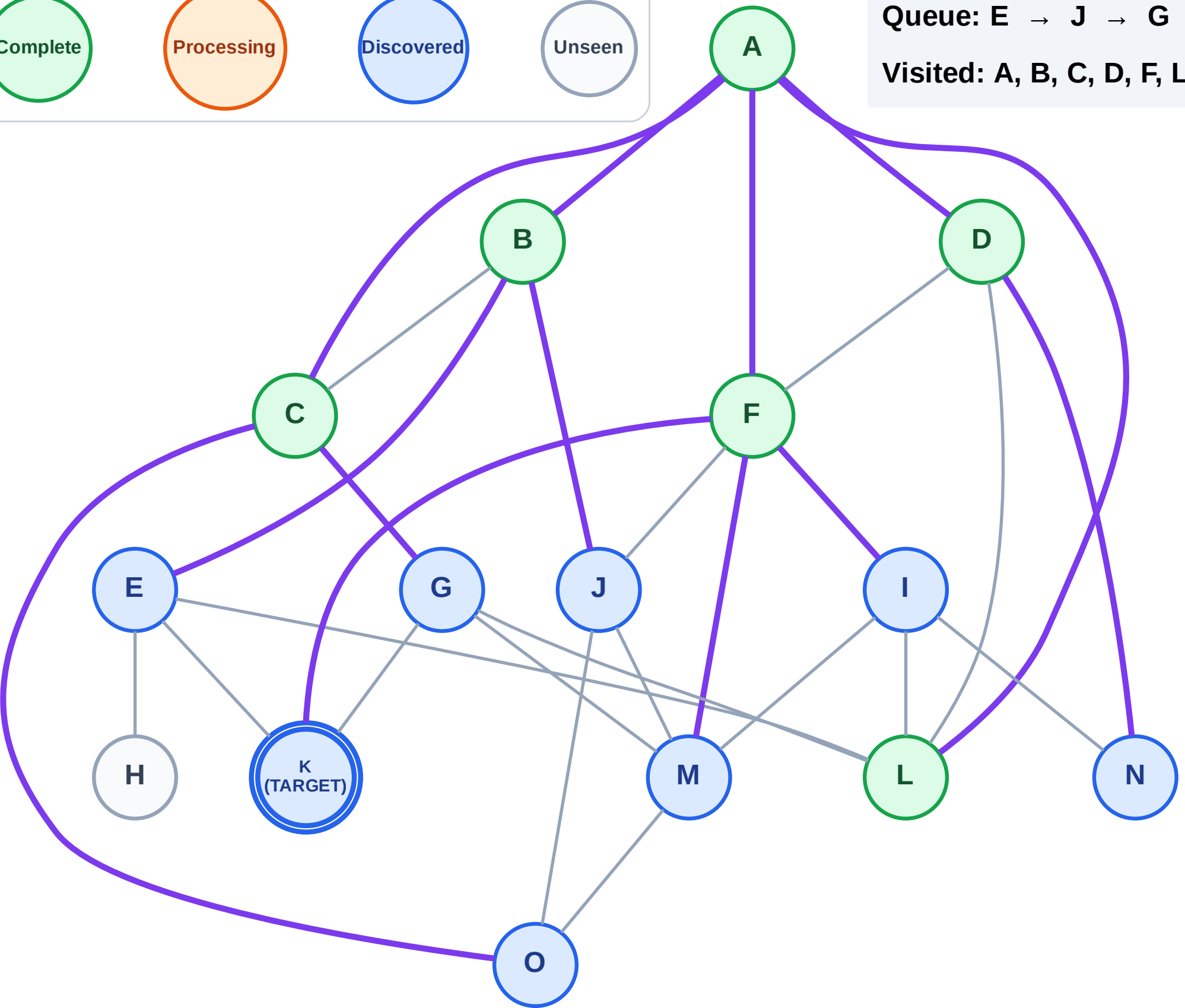
Processing

Discovered

Unseen

Queue: E → J → G → O → N → I → K → M

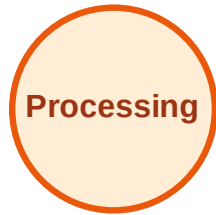
Visited: A, B, C, D, F, L



BFS Traversal

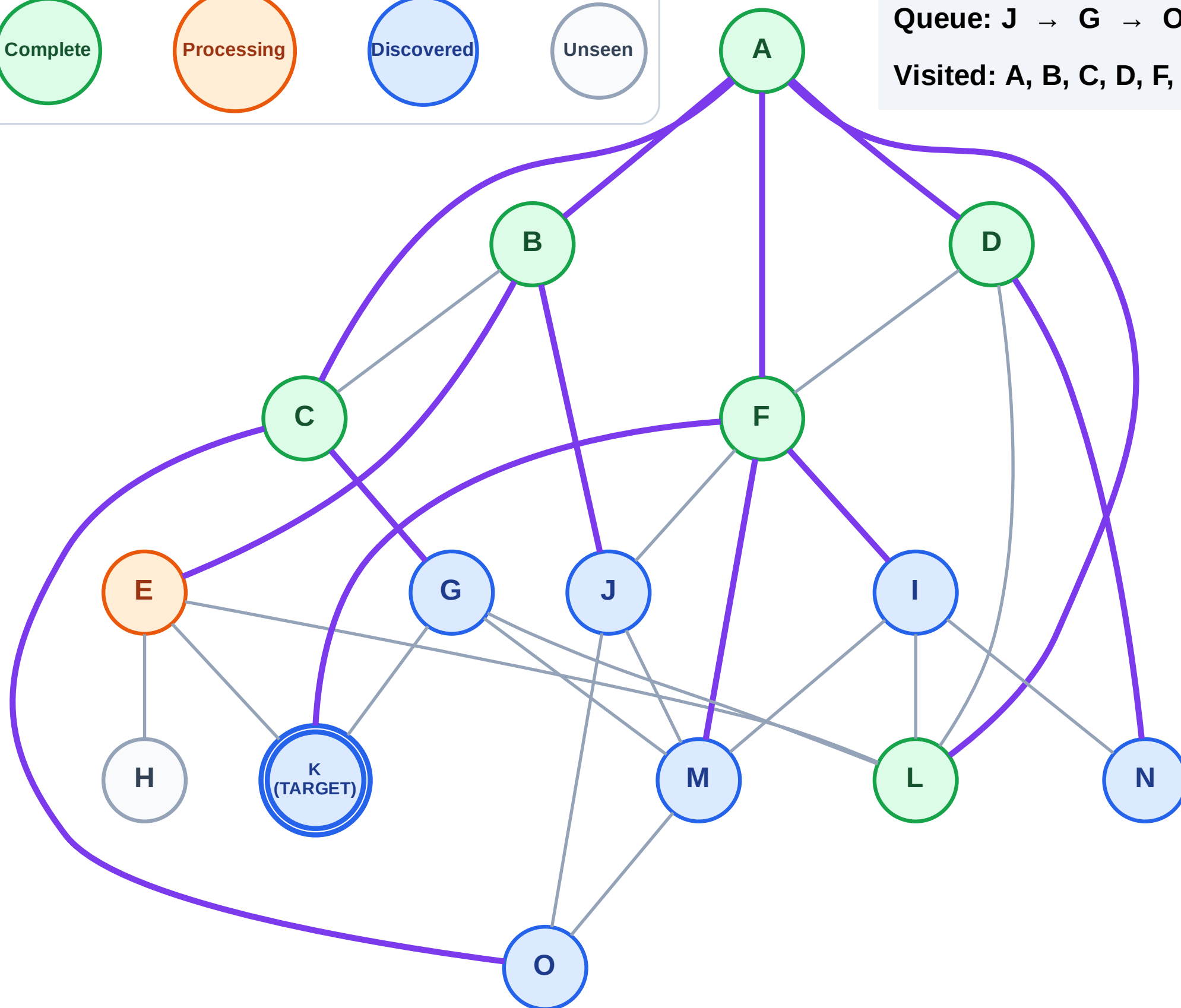
Step 14: Process node E

Legend



Queue: J → G → O → N → I → K → M

Visited: A, B, C, D, F, L



BFS Traversal

Step 15: Discover H from E

Legend

Complete

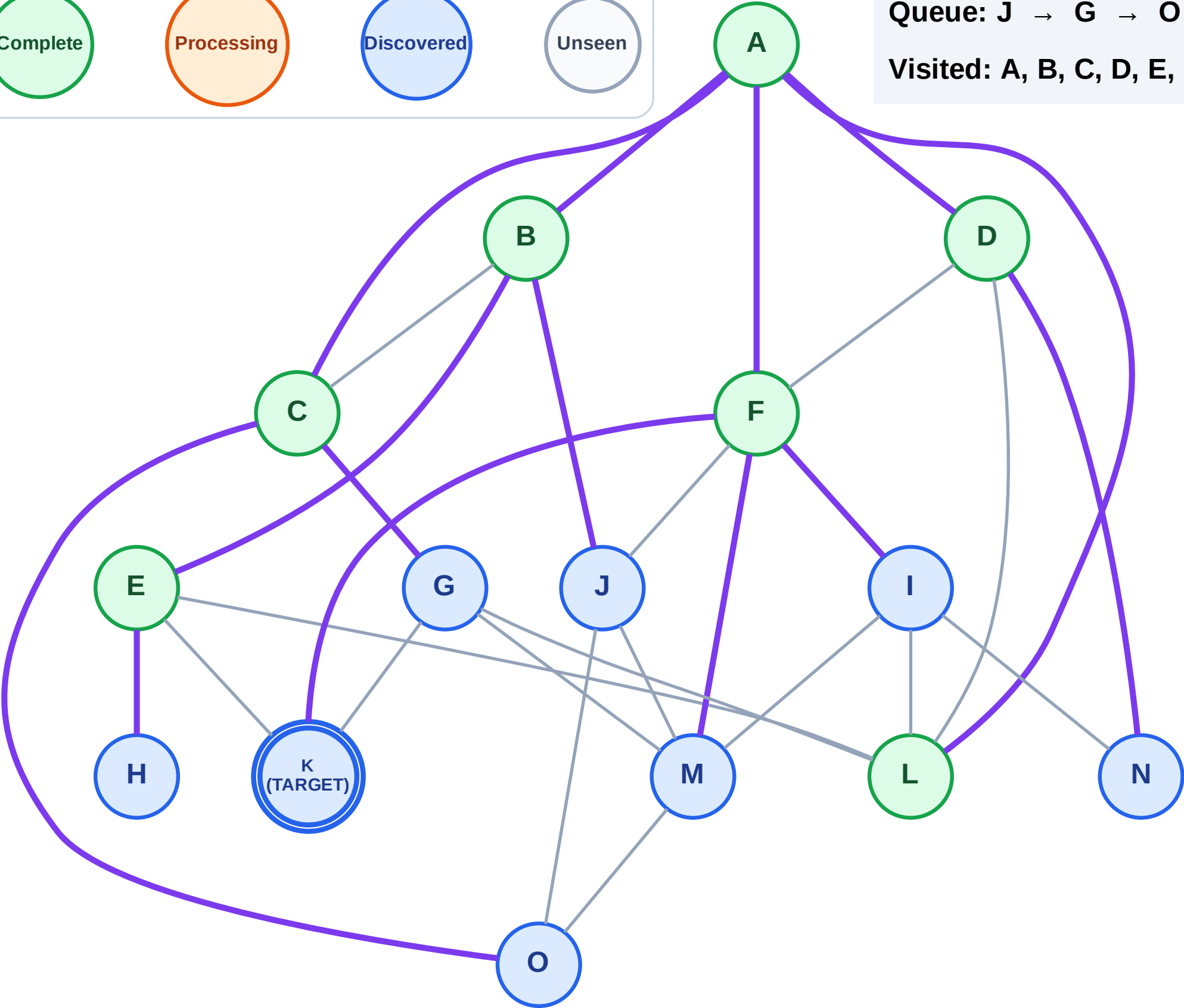
Processing

Discovered

Unseen

Queue: J → G → O → N → I → K → M → H

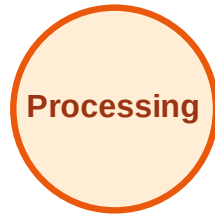
Visited: A, B, C, D, E, F, L



BFS Traversal

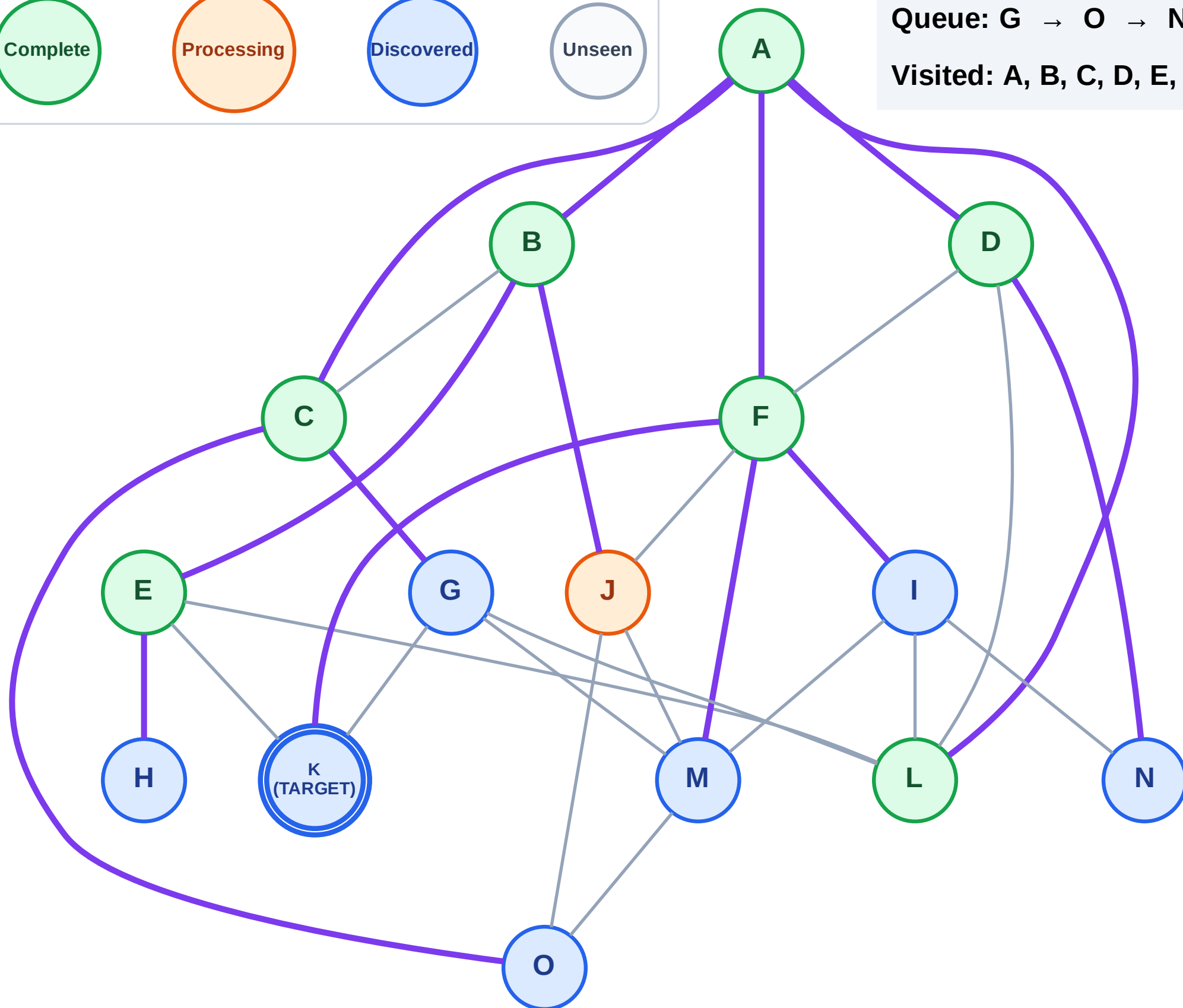
Step 16: Process node J

Legend



Queue: G → O → N → I → K → M → H

Visited: A, B, C, D, E, F, L

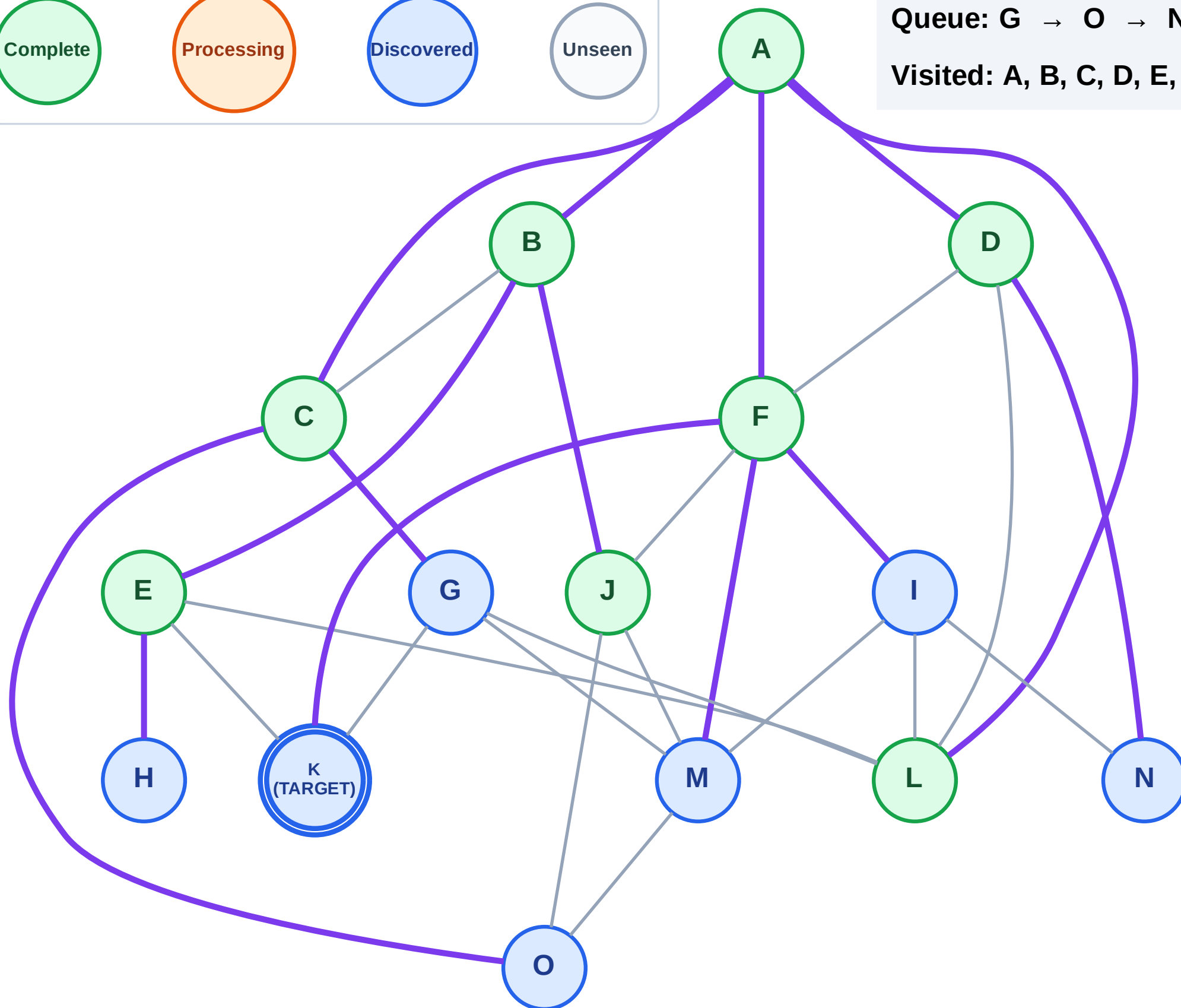


Step 17: No new neighbors from J

Legend

Unseen

Visited: A, B, C, D, E, F, J, L



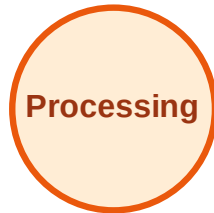
BFS Traversal

Step 18: Process node G

Legend



Complete



Processing



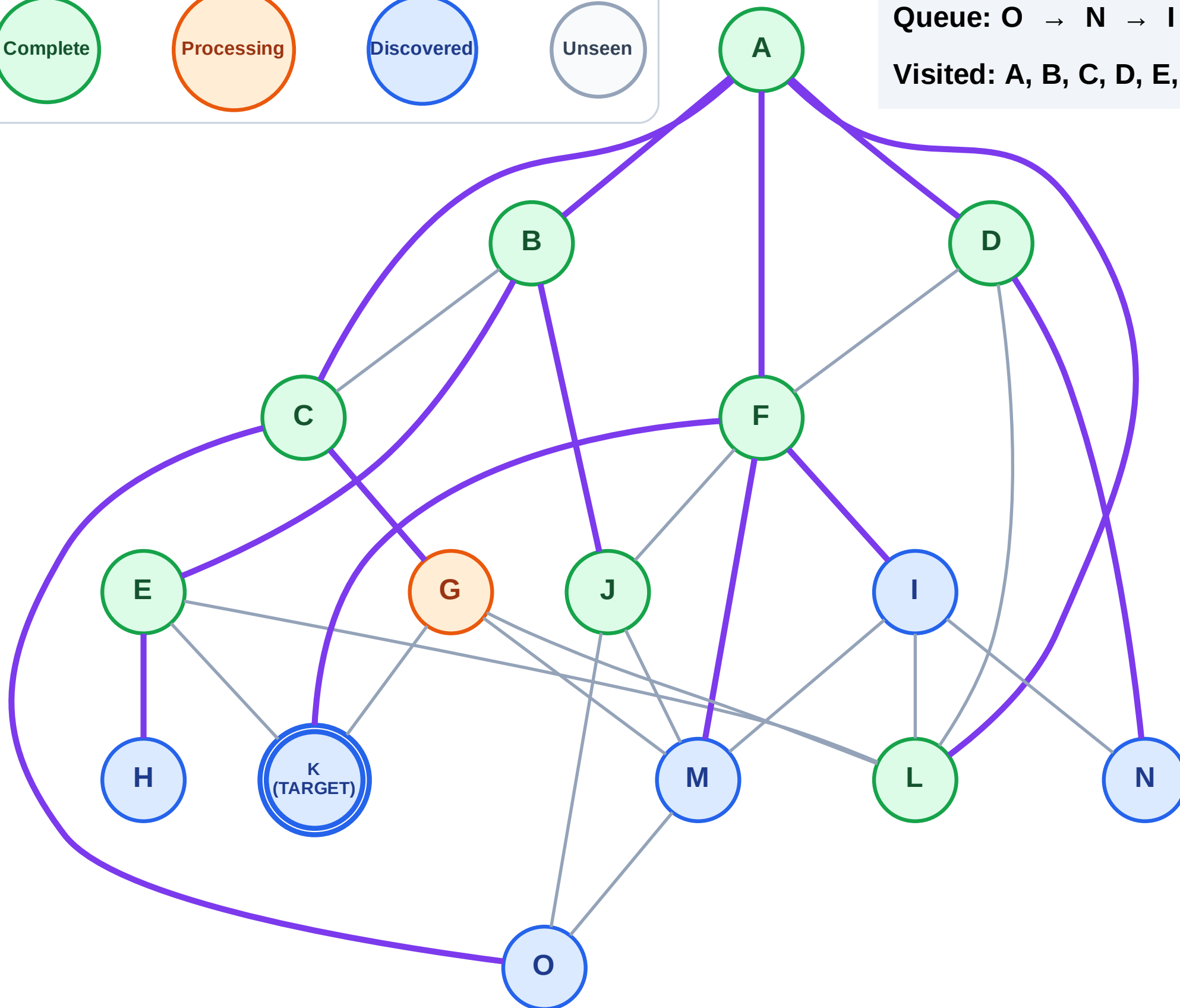
Discovered



Unseen

Queue: O → N → I → K → M → H

Visited: A, B, C, D, E, F, J, L



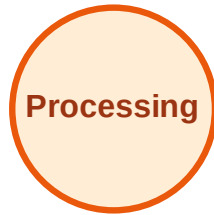
BFS Traversal

Step 19: No new neighbors from G

Legend



Complete



Processing



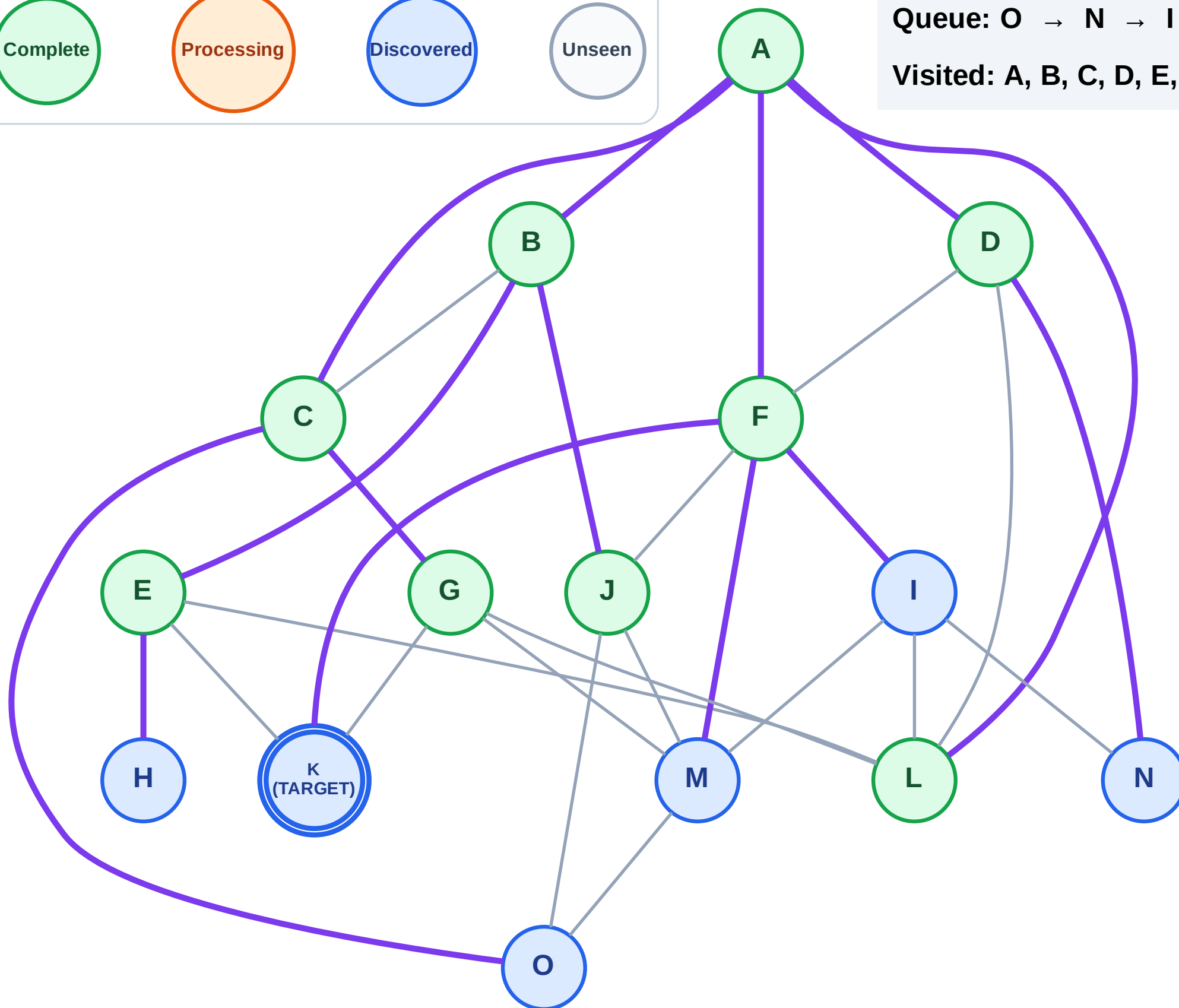
Discovered



Unseen

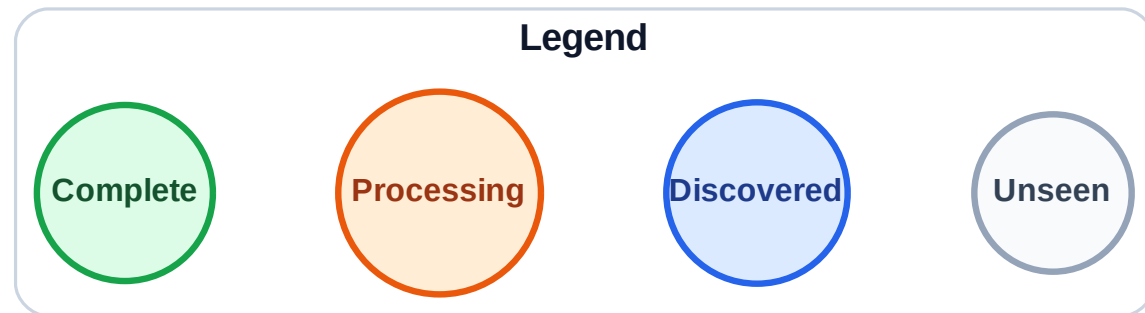
Queue: O → N → I → K → M → H

Visited: A, B, C, D, E, F, G, J, L

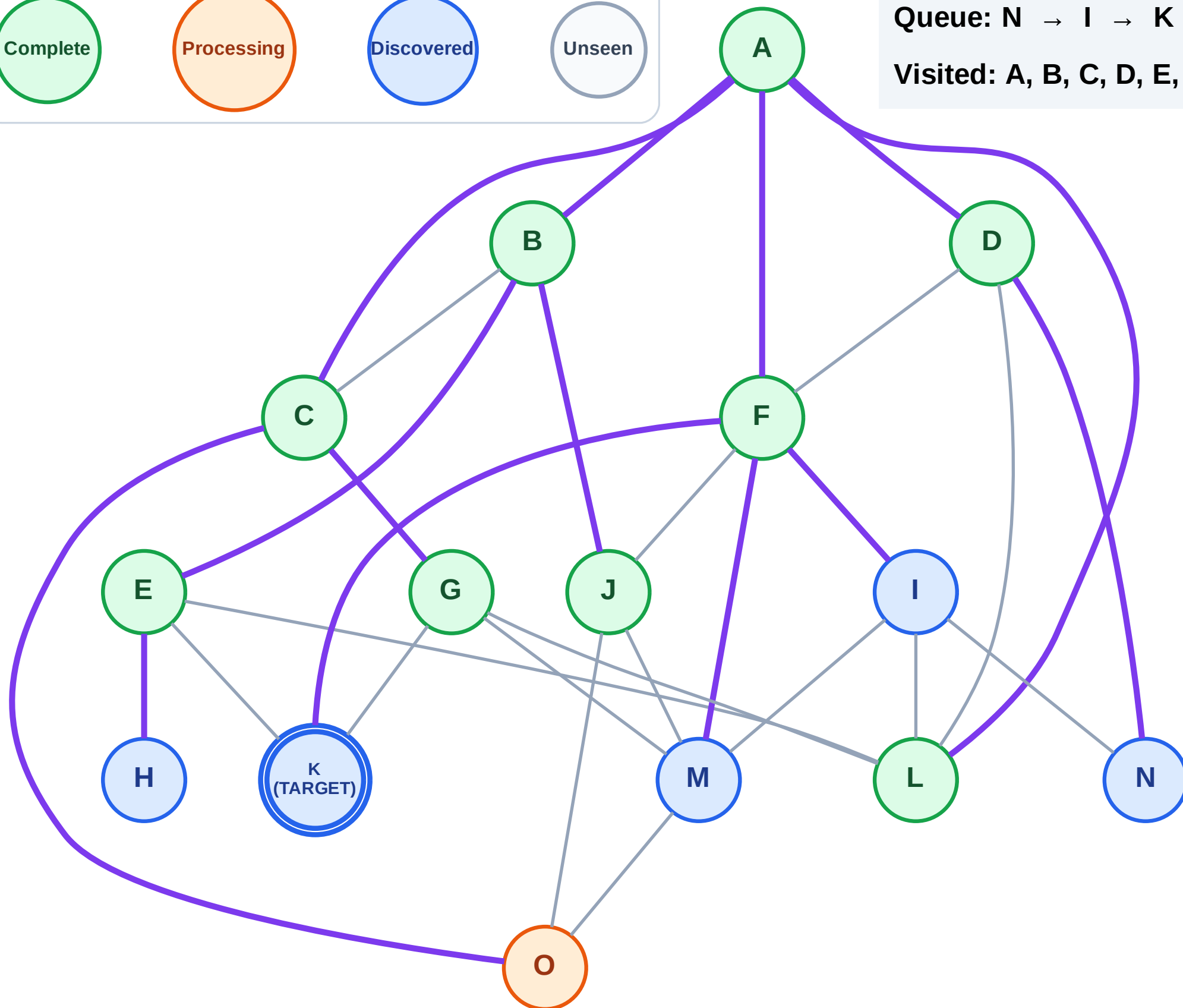


BFS Traversal

Step 20: Process node O

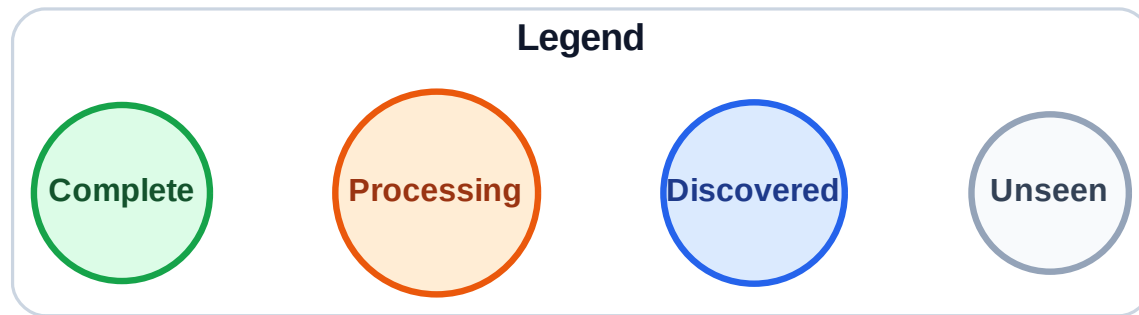


Queue: N → I → K → M → H
Visited: A, B, C, D, E, F, G, J, L



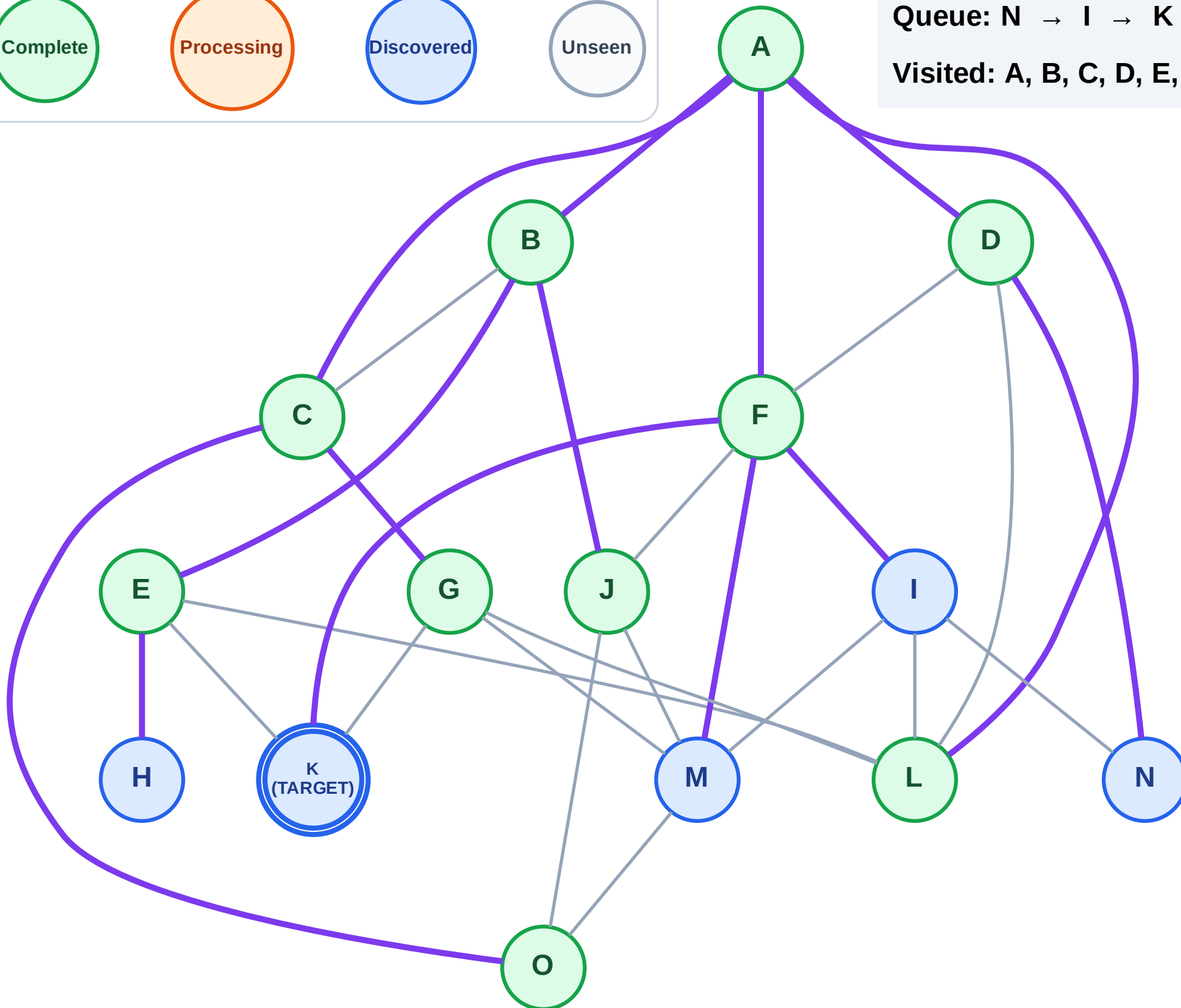
BFS Traversal

Step 21: No new neighbors from O



Queue: N → I → K → M → H

Visited: A, B, C, D, E, F, G, J, L, O



BFS Traversal

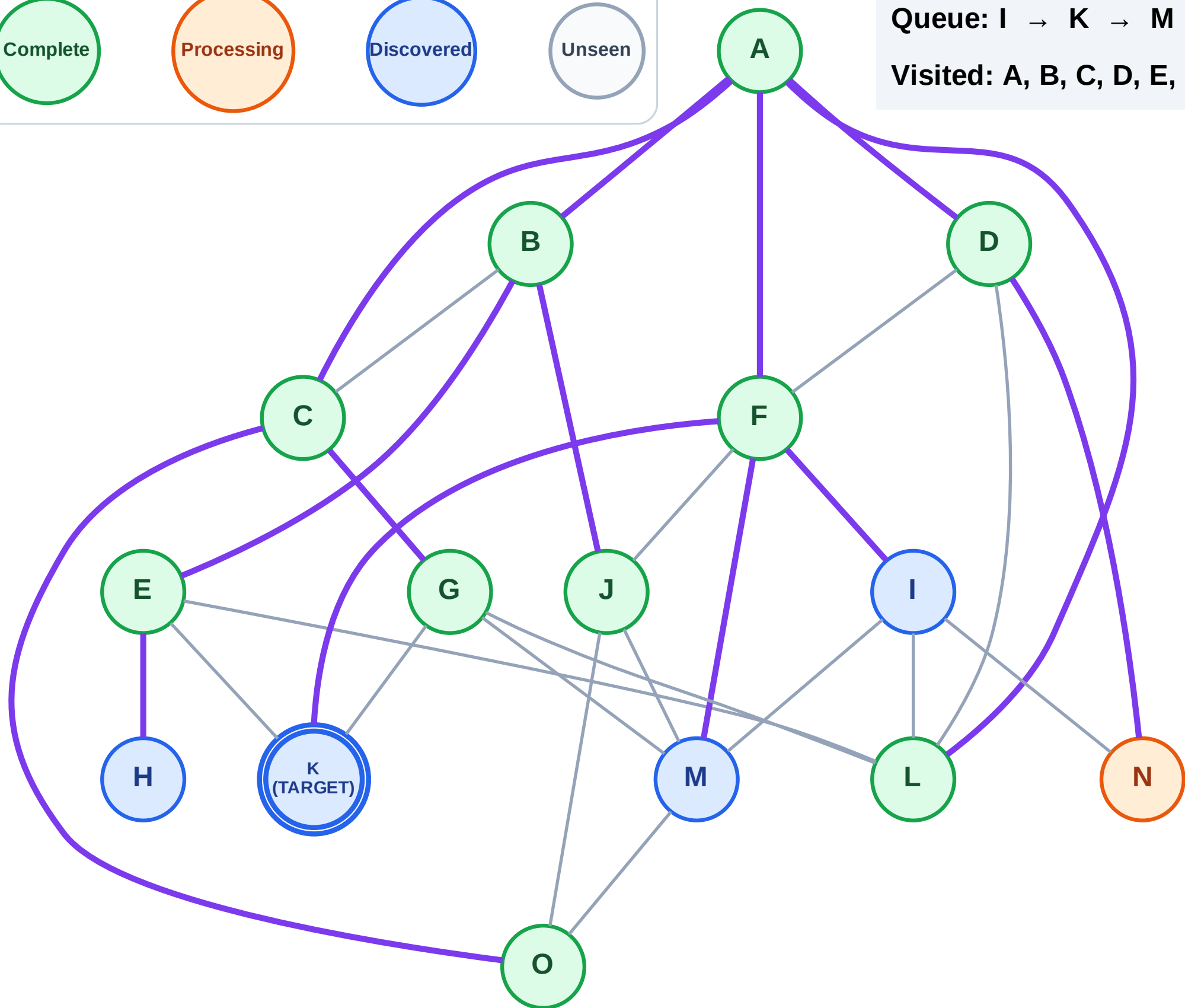
Step 22: Process node N

Legend



Queue: I → K → M → H

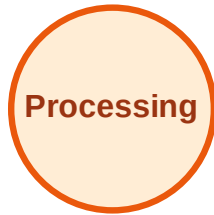
Visited: A, B, C, D, E, F, G, J, L, O



BFS Traversal

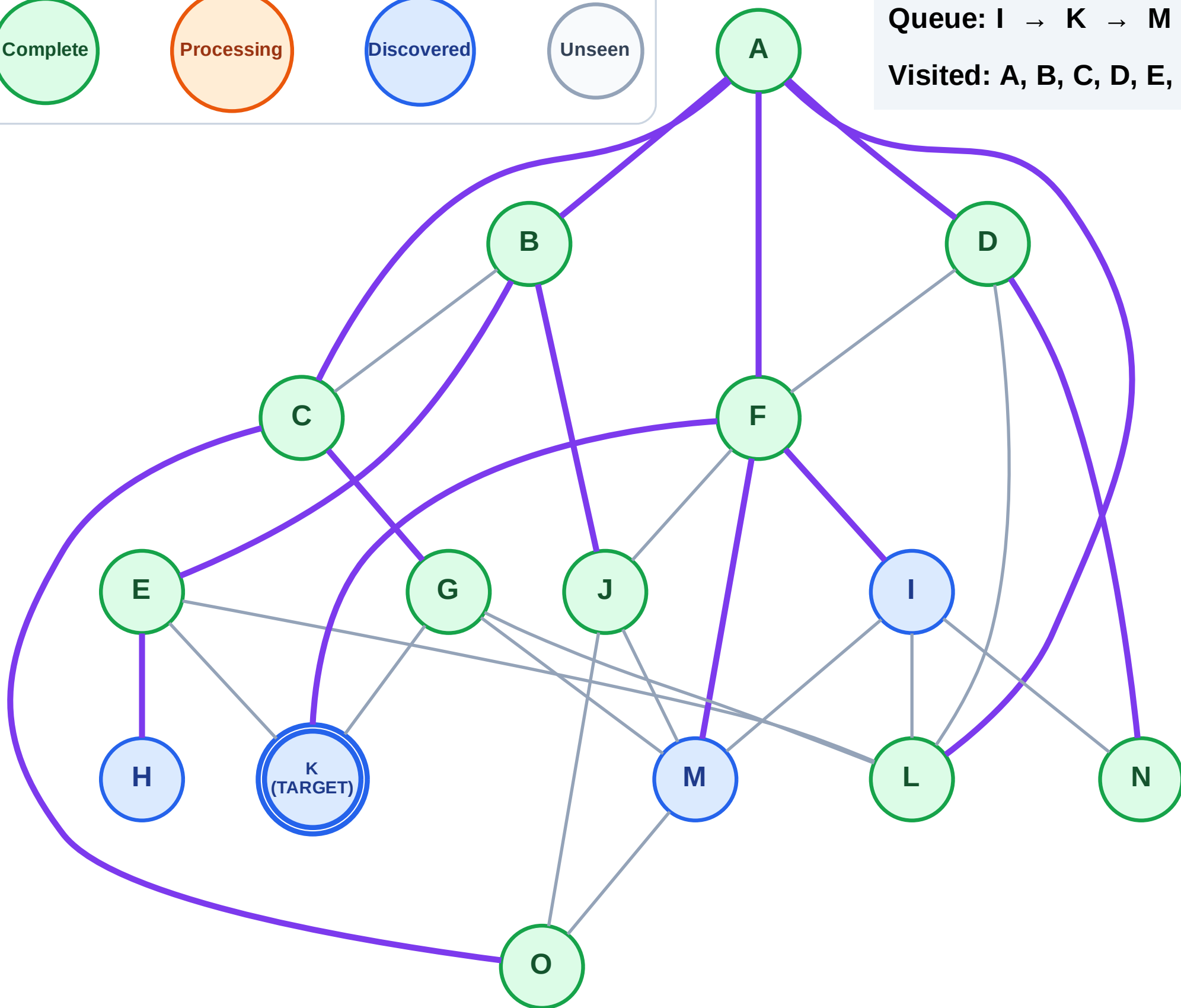
Step 23: No new neighbors from N

Legend



Queue: I → K → M → H

Visited: A, B, C, D, E, F, G, J, L, N, O



BFS Traversal

Step 24: Process node I

Legend

Complete

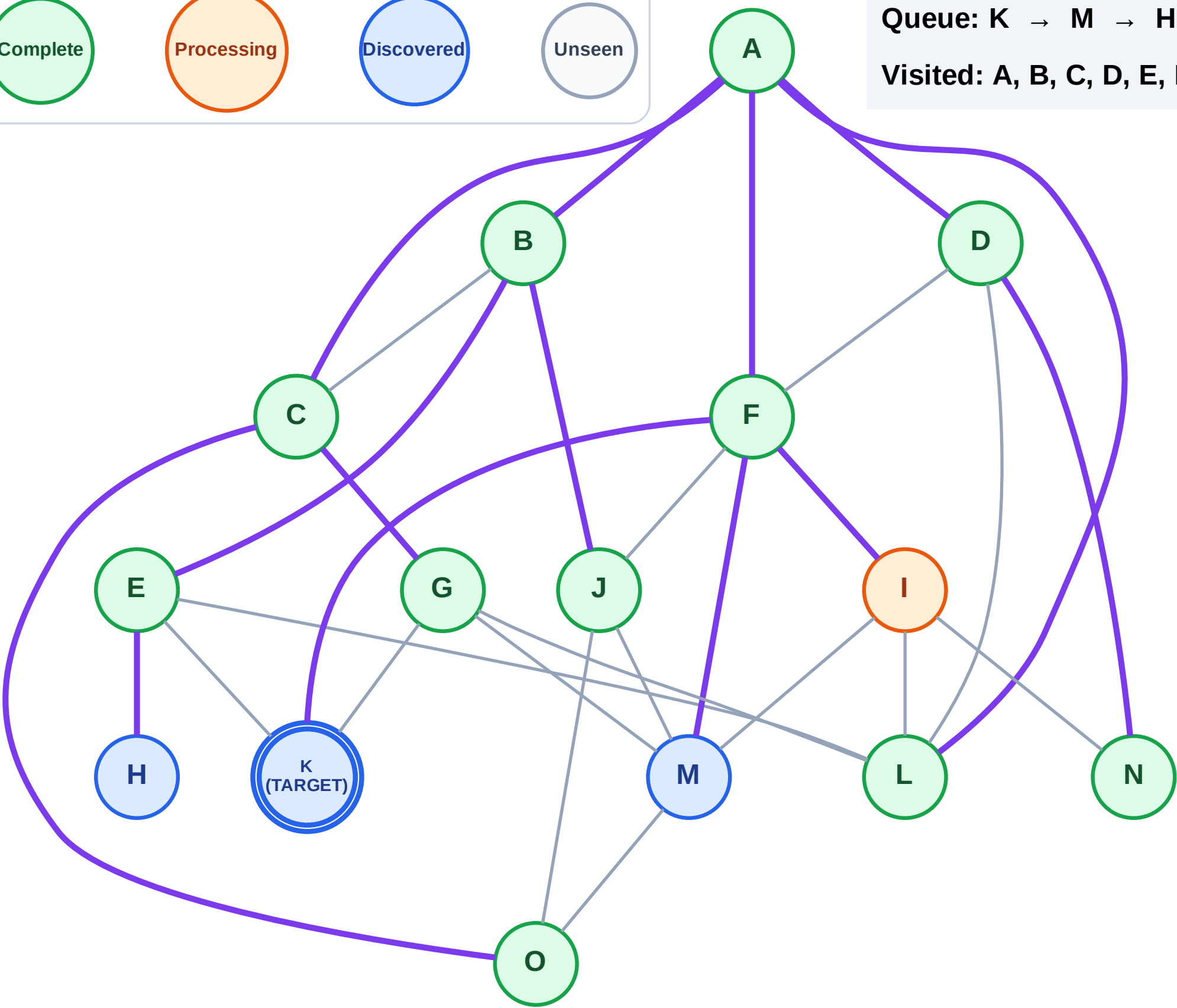
Processing

Discovered

Unseen

Queue: K → M → H

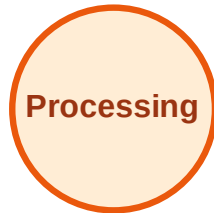
Visited: A, B, C, D, E, F, G, J, L, N, O



BFS Traversal

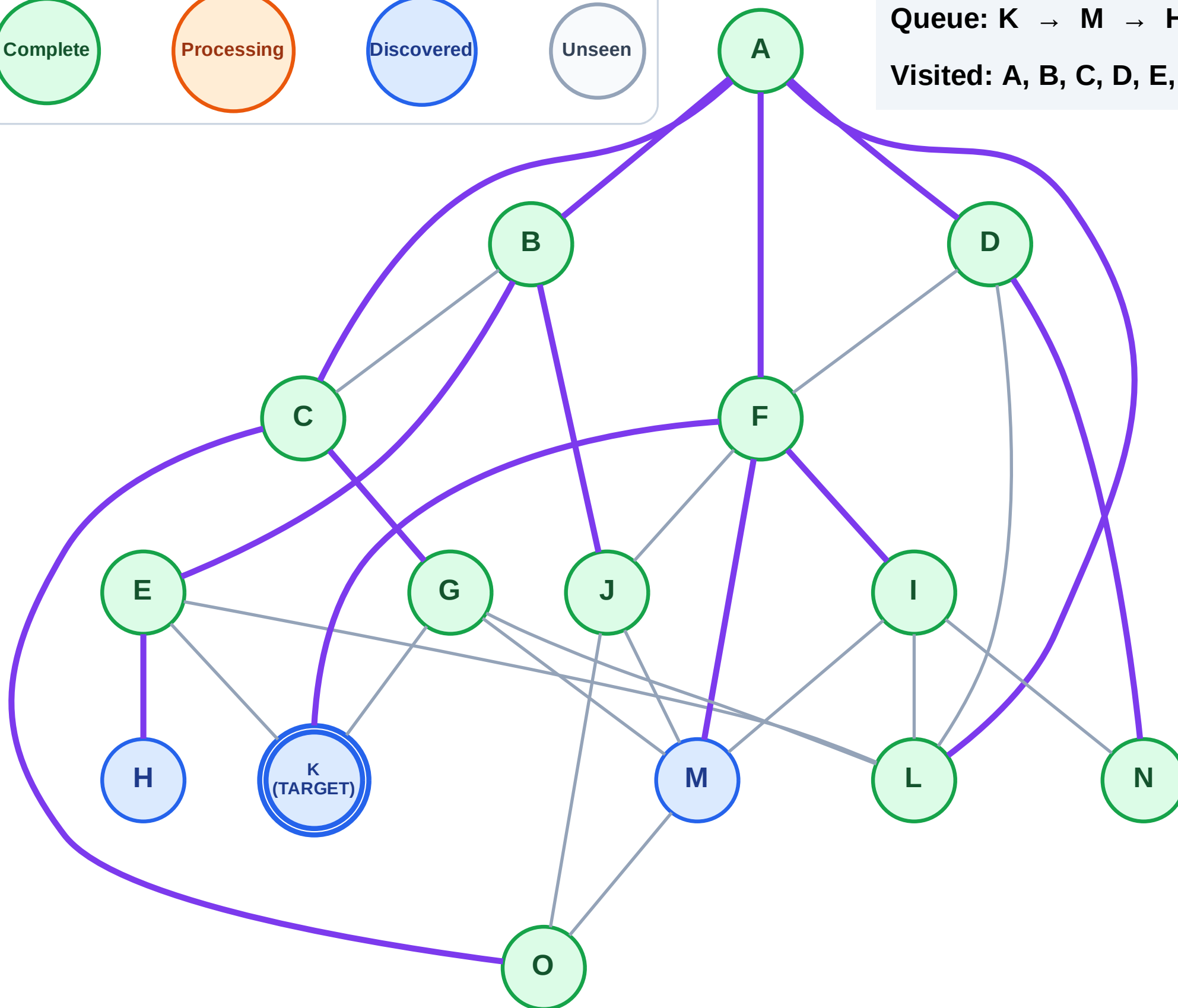
Step 25: No new neighbors from I

Legend



Queue: K → M → H

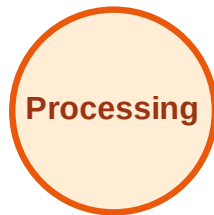
Visited: A, B, C, D, E, F, G, I, J, L, N, O



BFS Traversal

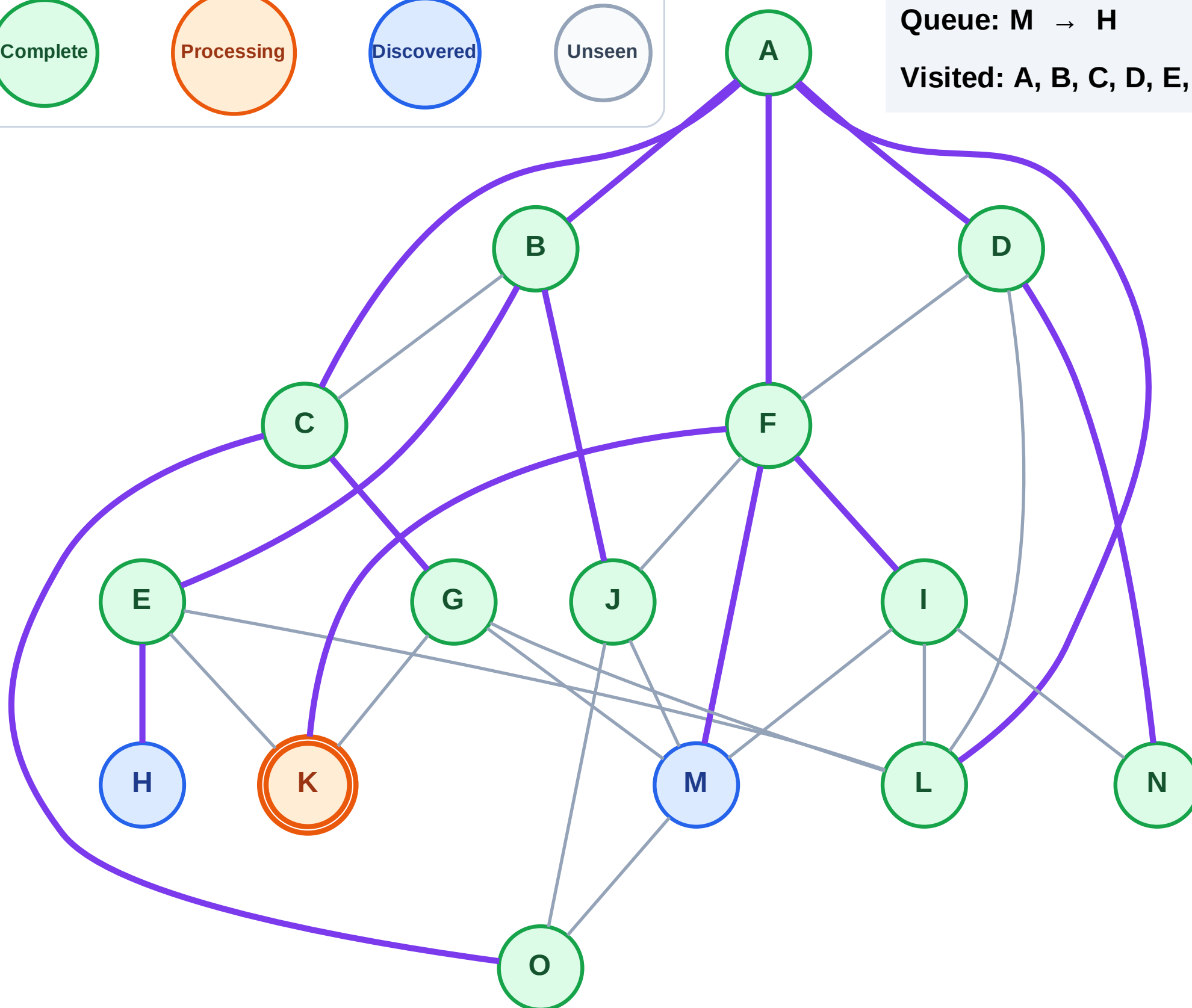
Step 26: Process node K

Legend



Queue: M → H

Visited: A, B, C, D, E, F, G, I, J, L, N, O



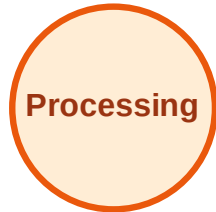
BFS Traversal

Step 27: TARGET NODE K REACHED! Path (2 hops): A → F → K

Legend



Complete



Processing



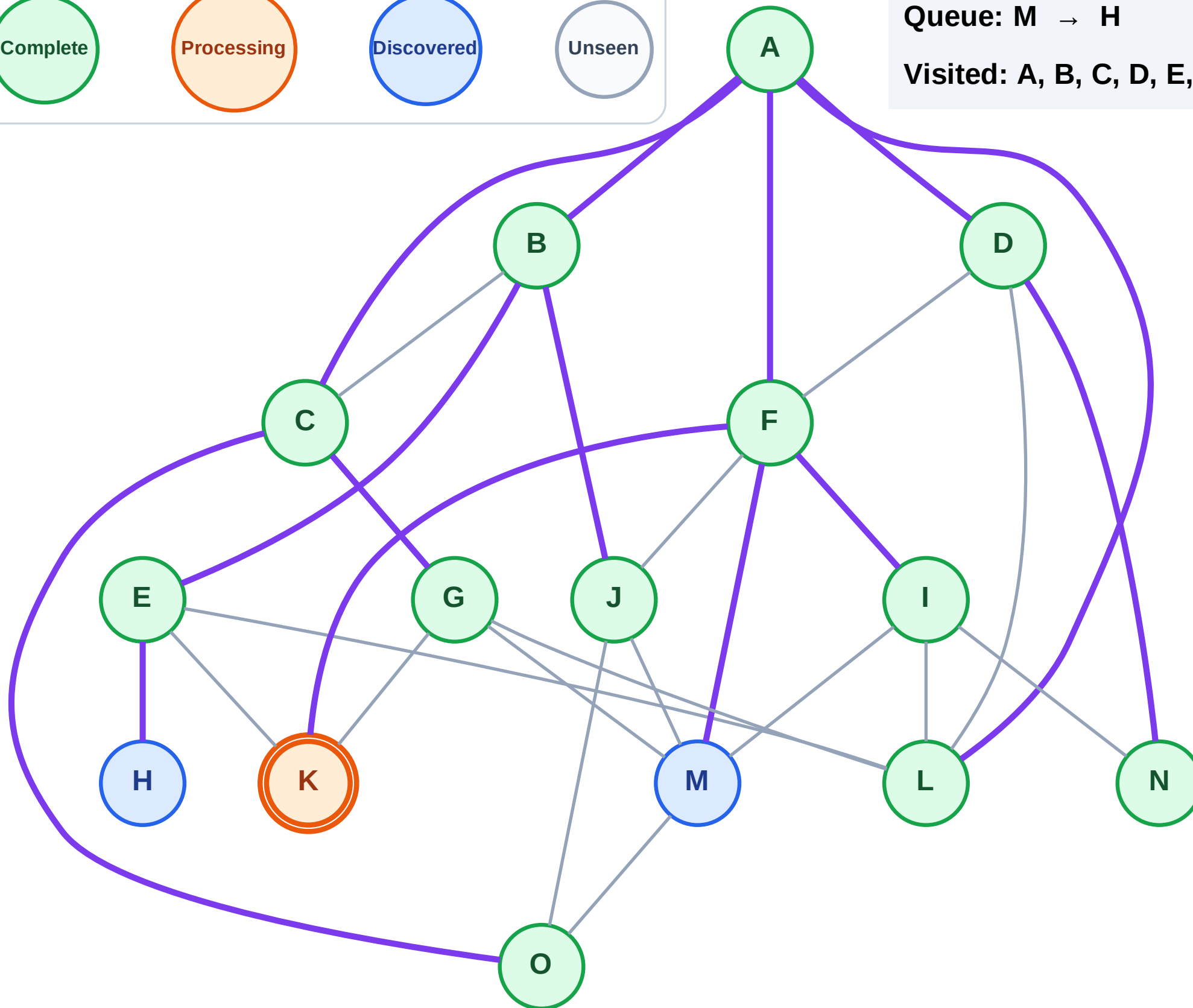
Discovered



Unseen

Queue: M → H

Visited: A, B, C, D, E, F, G, I, J, K, L, N, O



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

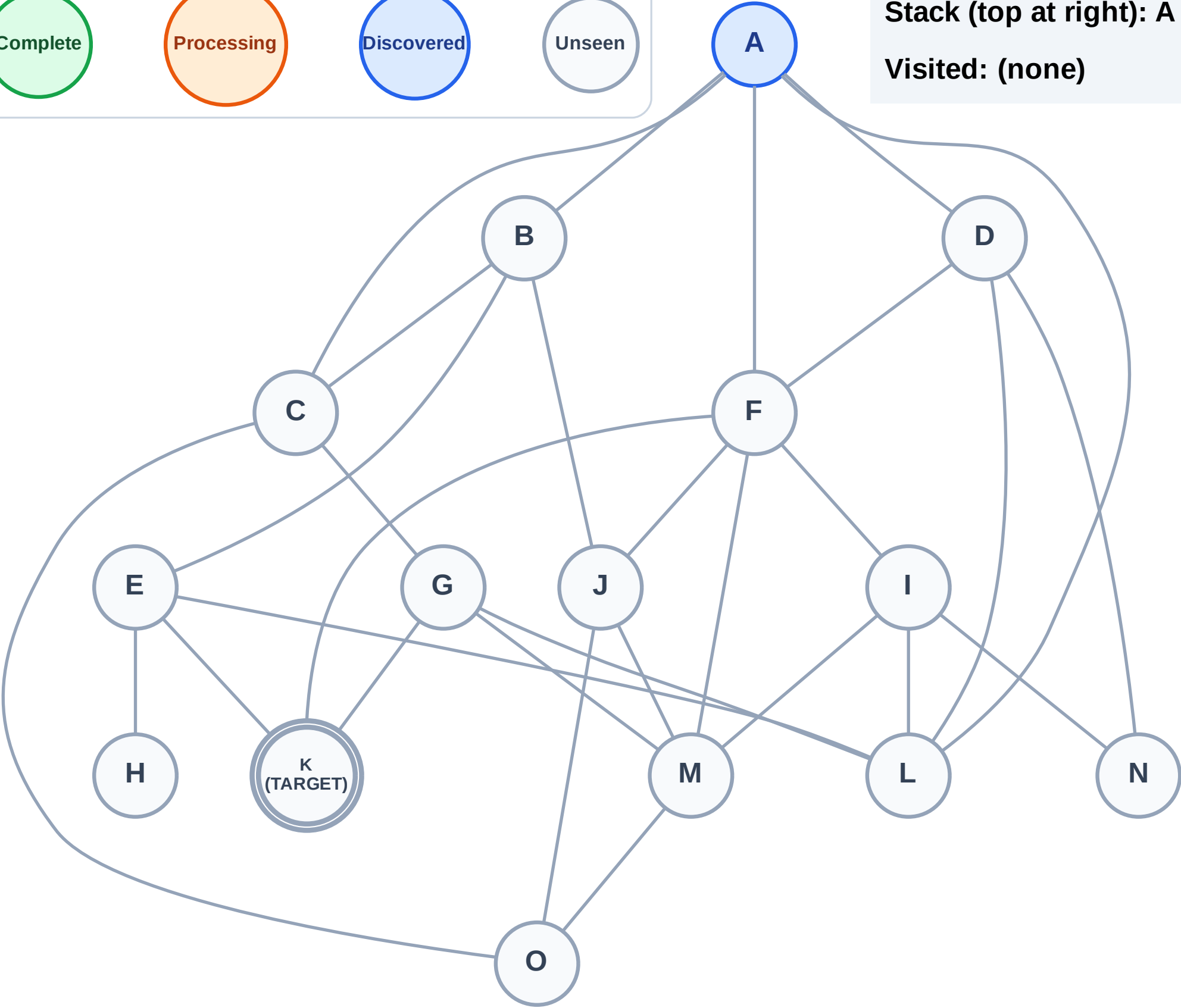
Complete

Processing

Discovered

Unseen

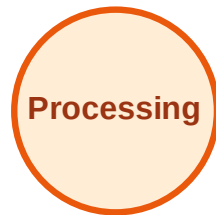
Stack (top at right): A
Visited: (none)



DFS Traversal

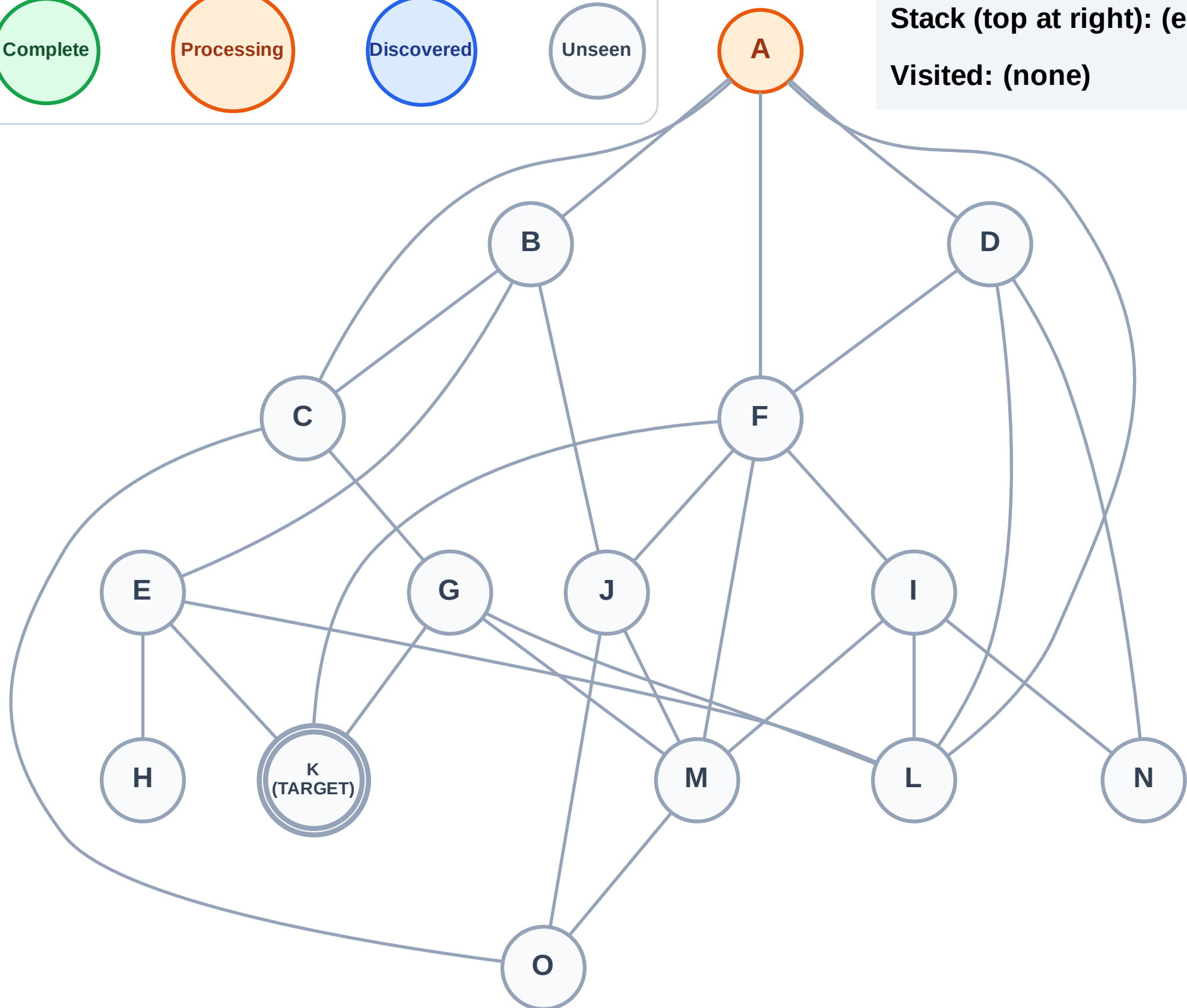
Step 2: Process node A

Legend



Stack (top at right): (empty)

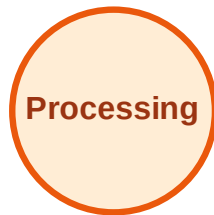
Visited: (none)



DFS Traversal

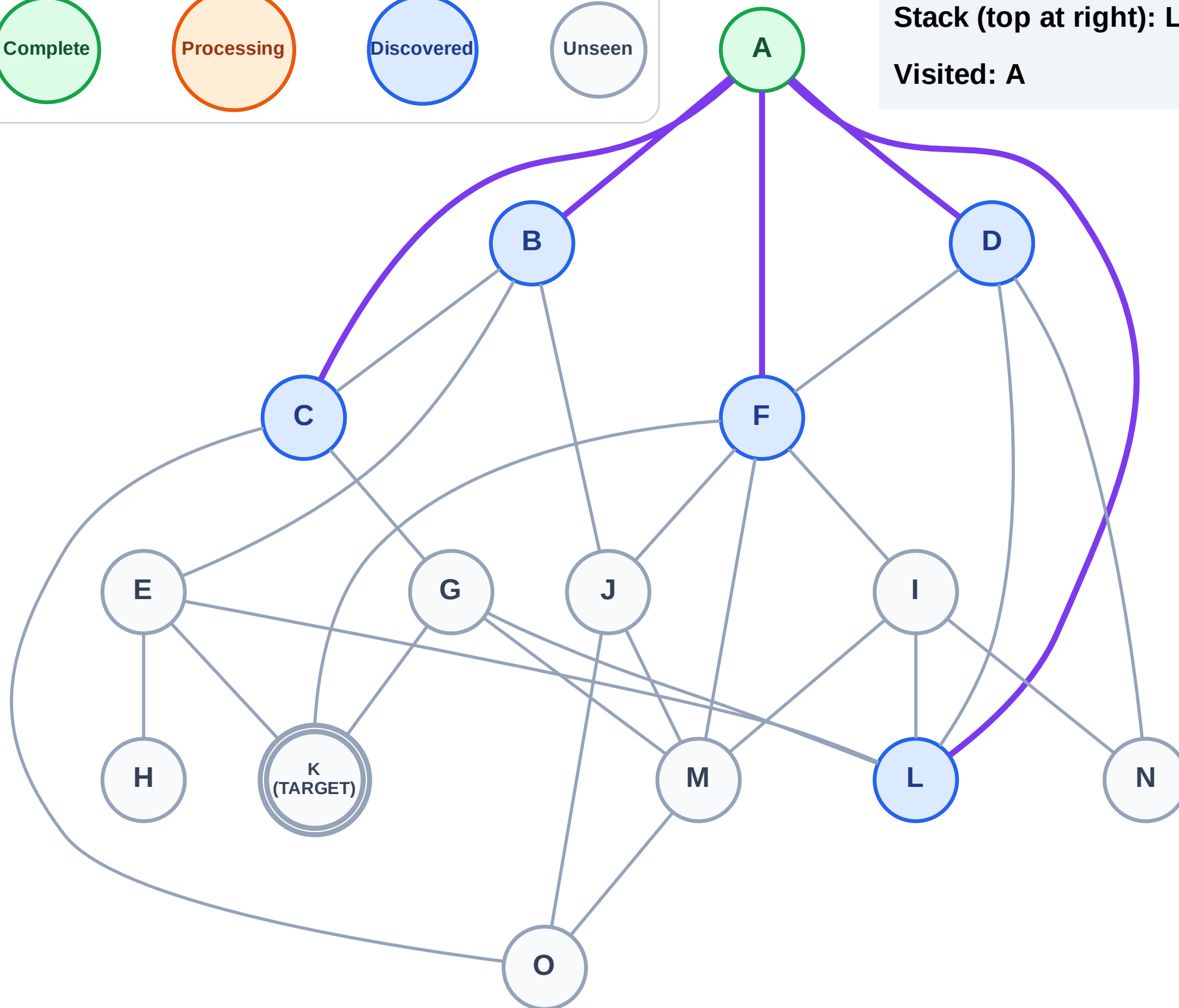
Step 3: Discover L, F, D, C, B from A

Legend



Stack (top at right): L | F | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

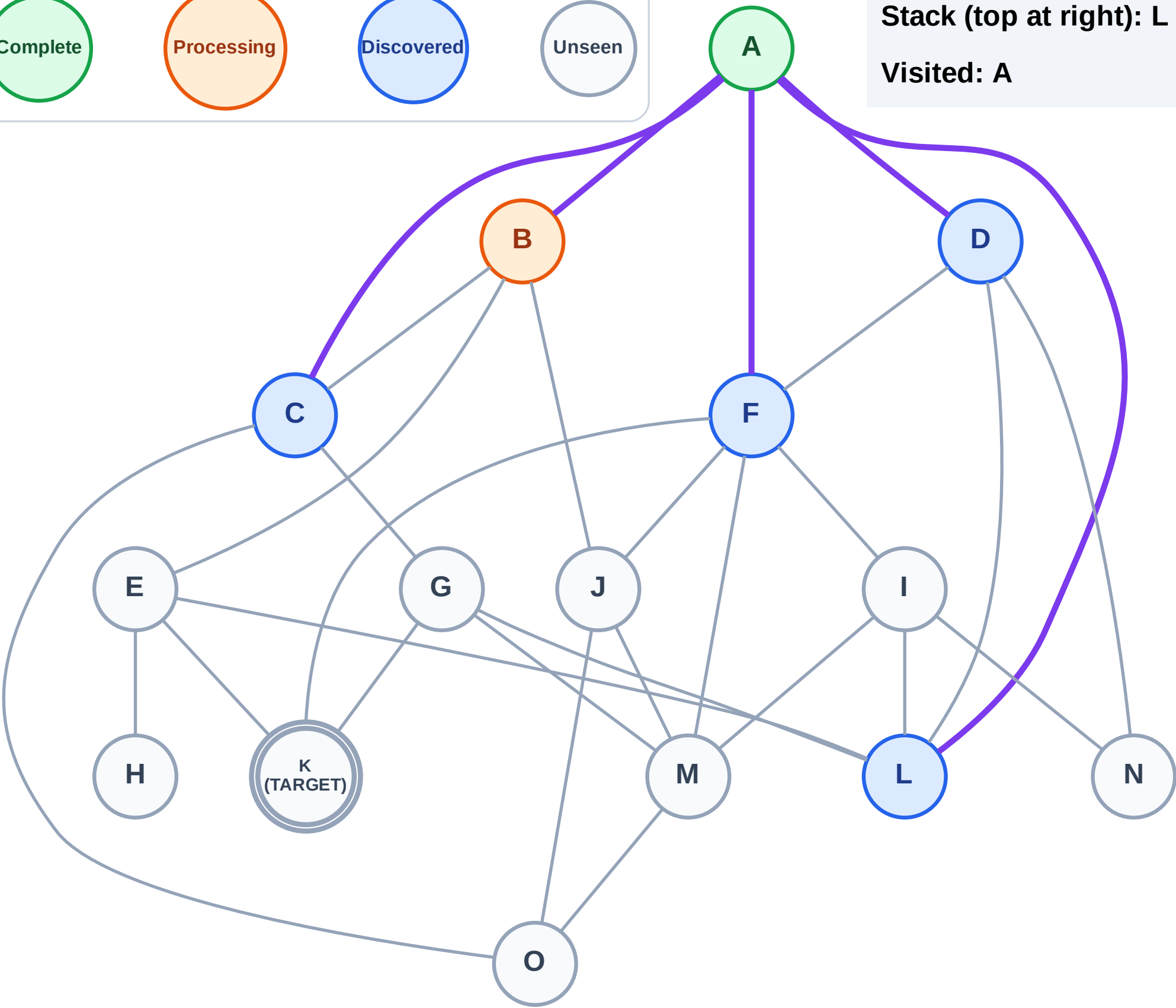
Complete

Processing

Discovered

Unseen

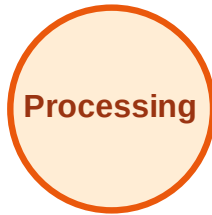
Stack (top at right): L | F | D | C
Visited: A



DFS Traversal

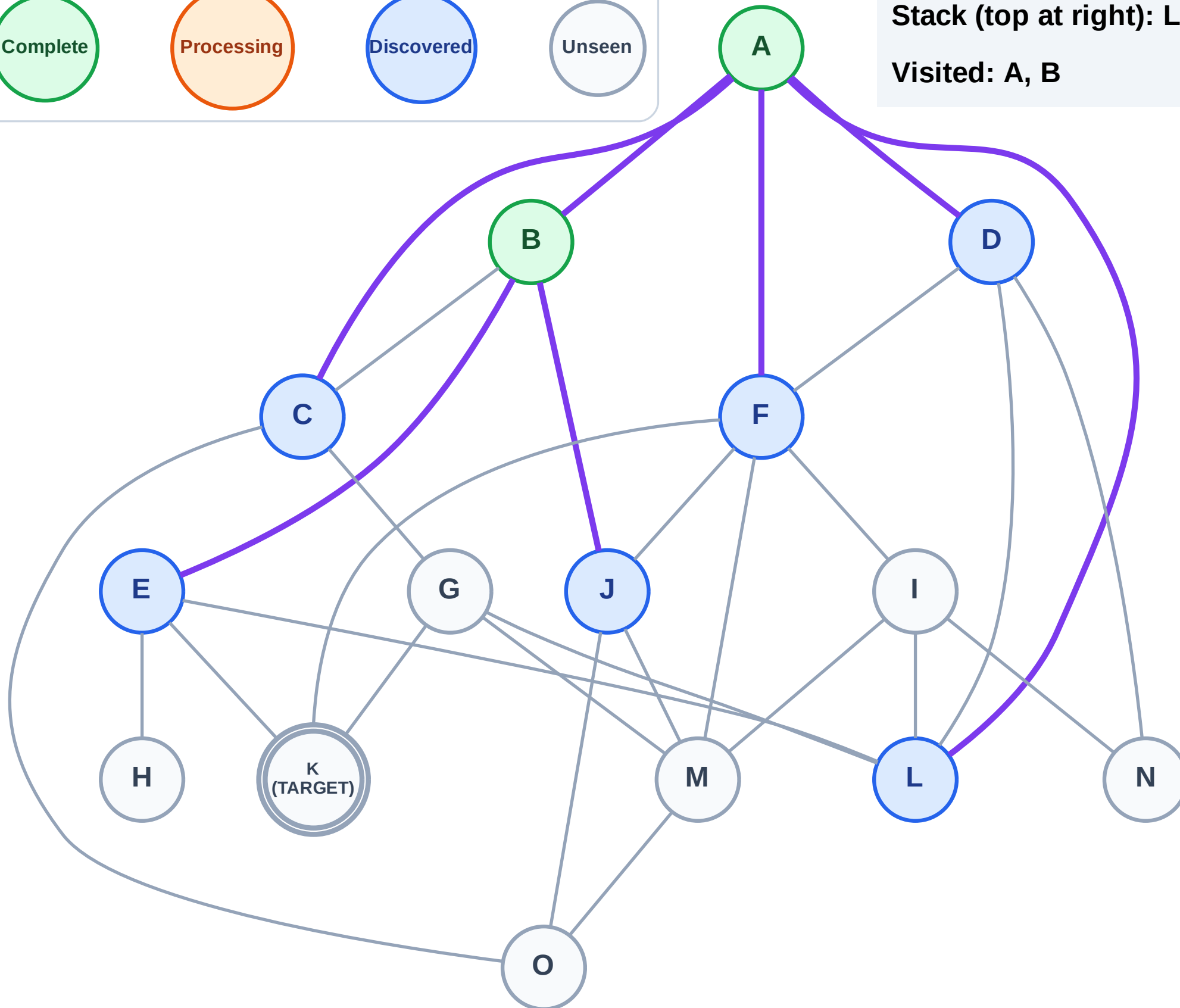
Step 5: Discover J, E from B

Legend



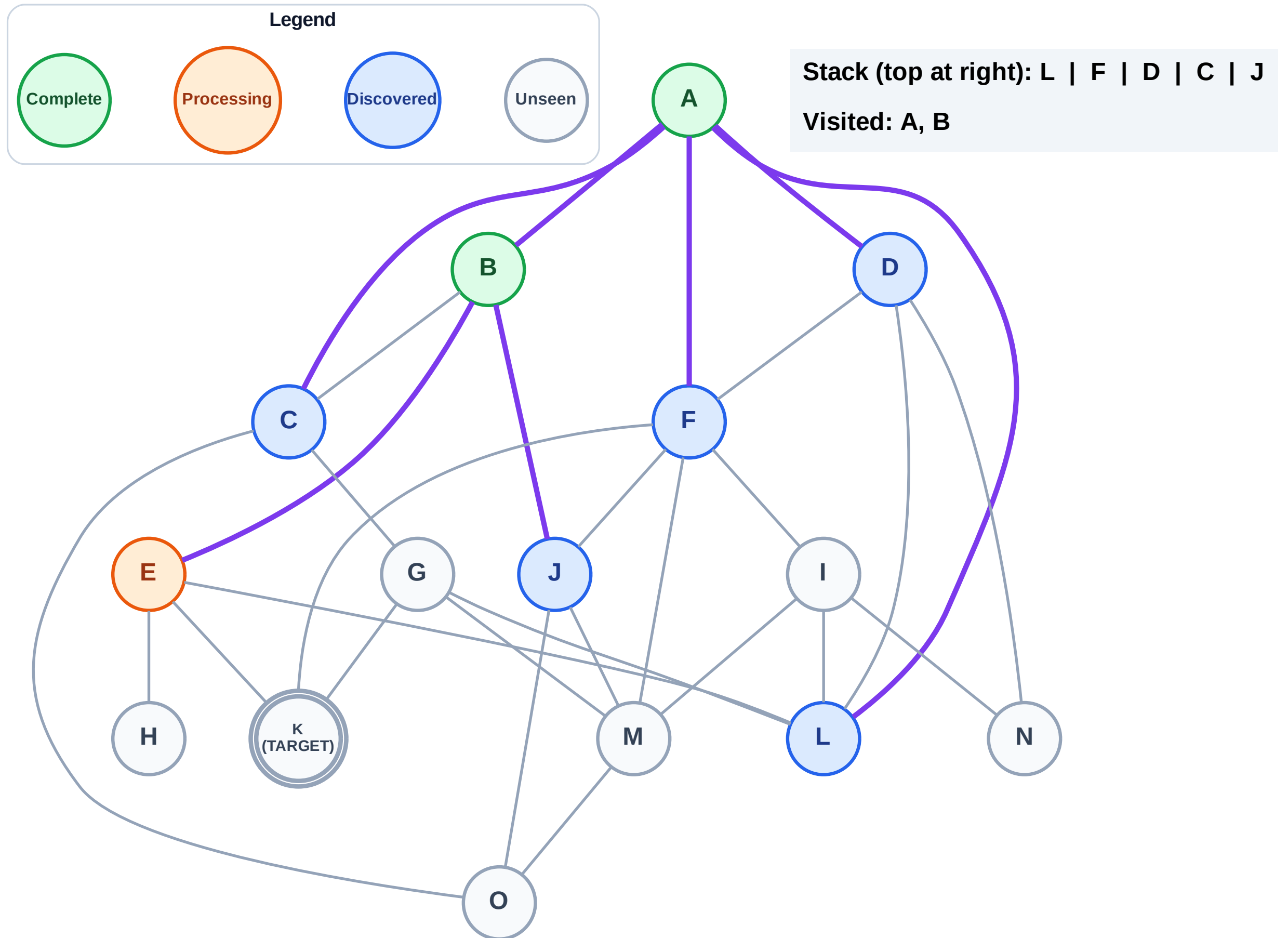
Stack (top at right): L | F | D | C | J | E

Visited: A, B



Step 6: Process node E

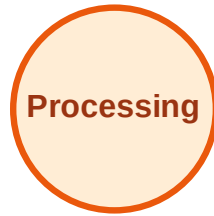
Step 6: Process node E



DFS Traversal

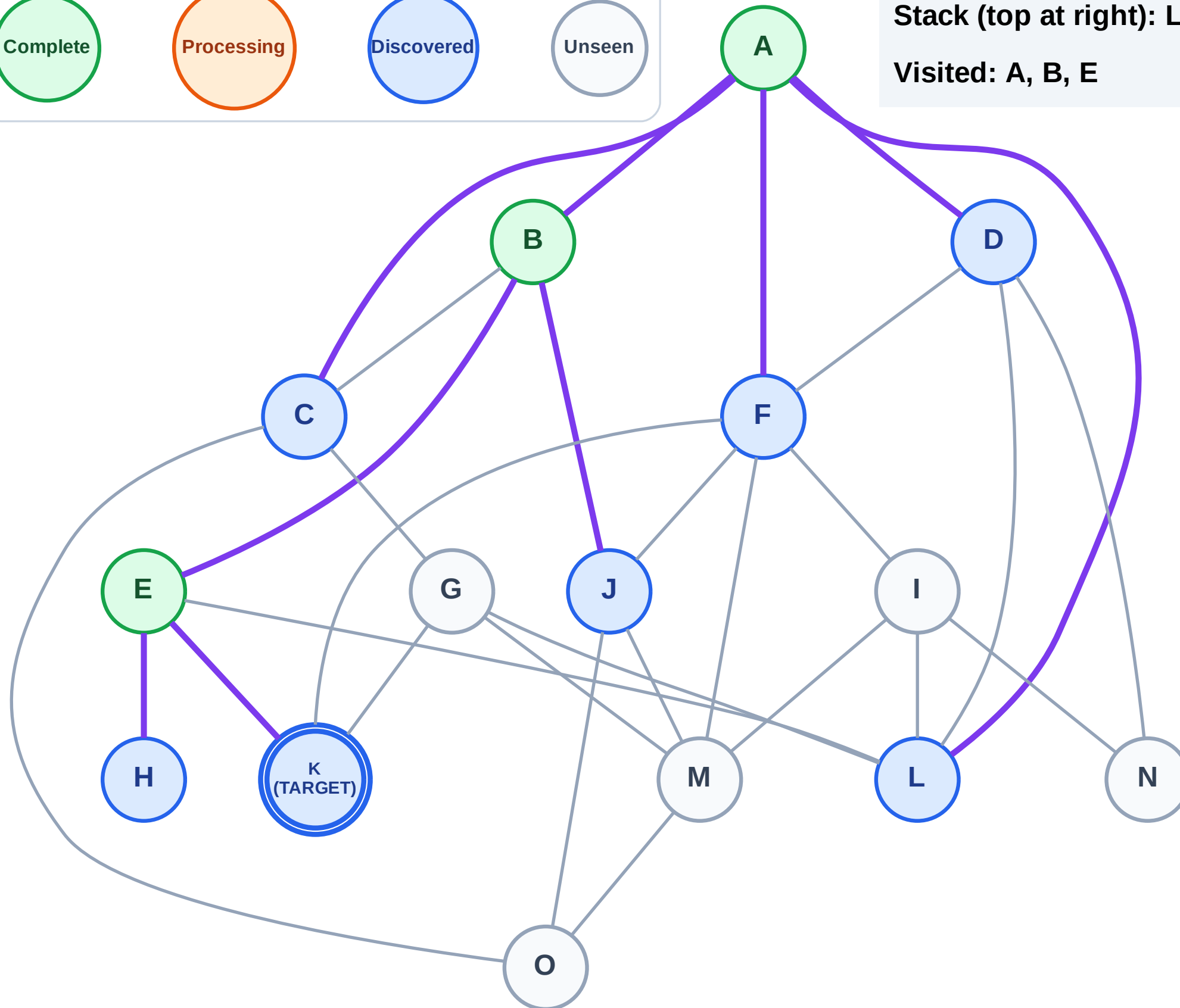
Step 7: Discover K, H from E

Legend

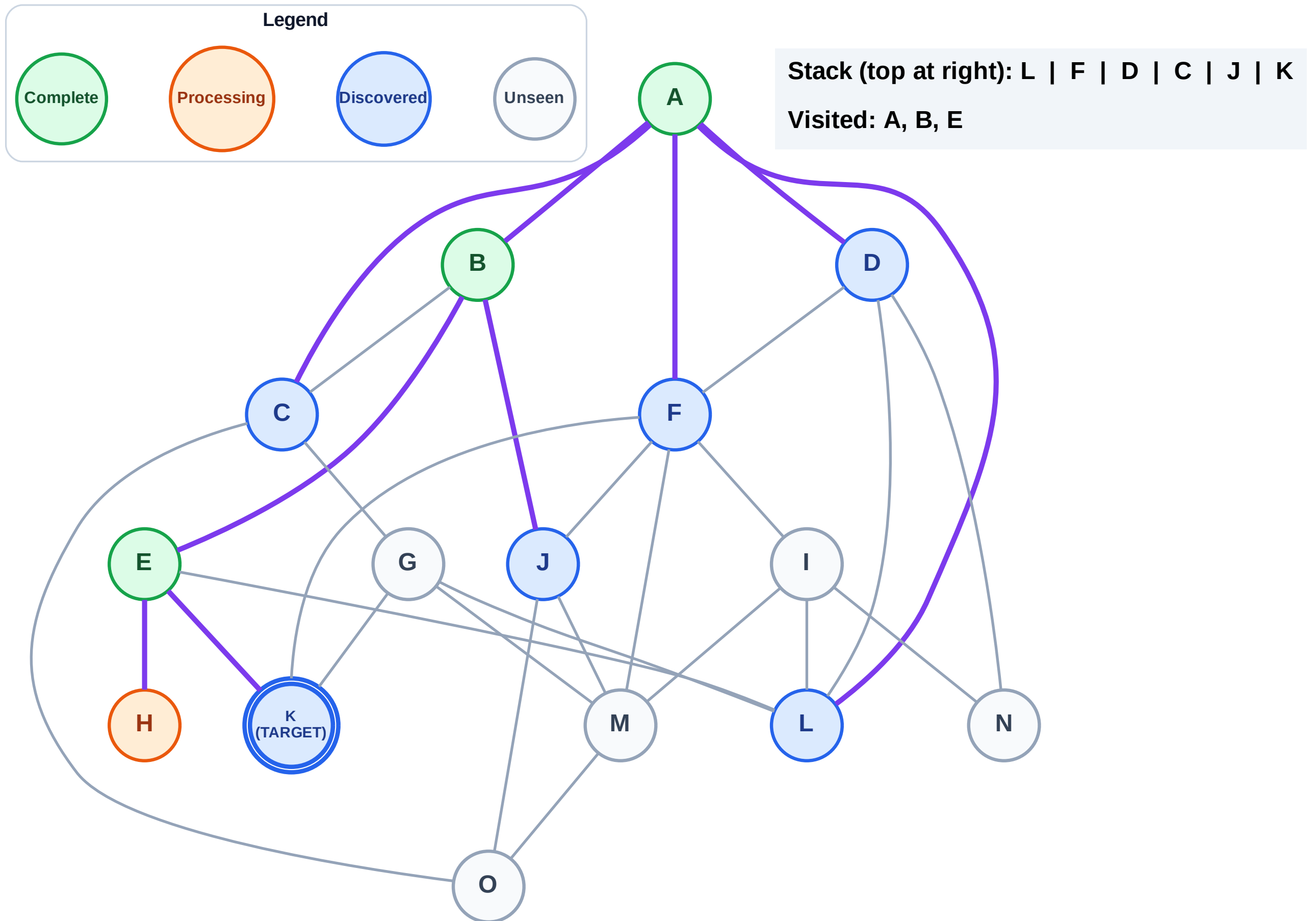


Stack (top at right): L | F | D | C | J | K | H

Visited: A, B, E



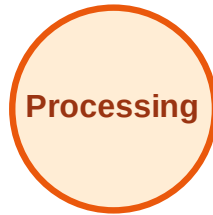
Step 8: Process node H



DFS Traversal

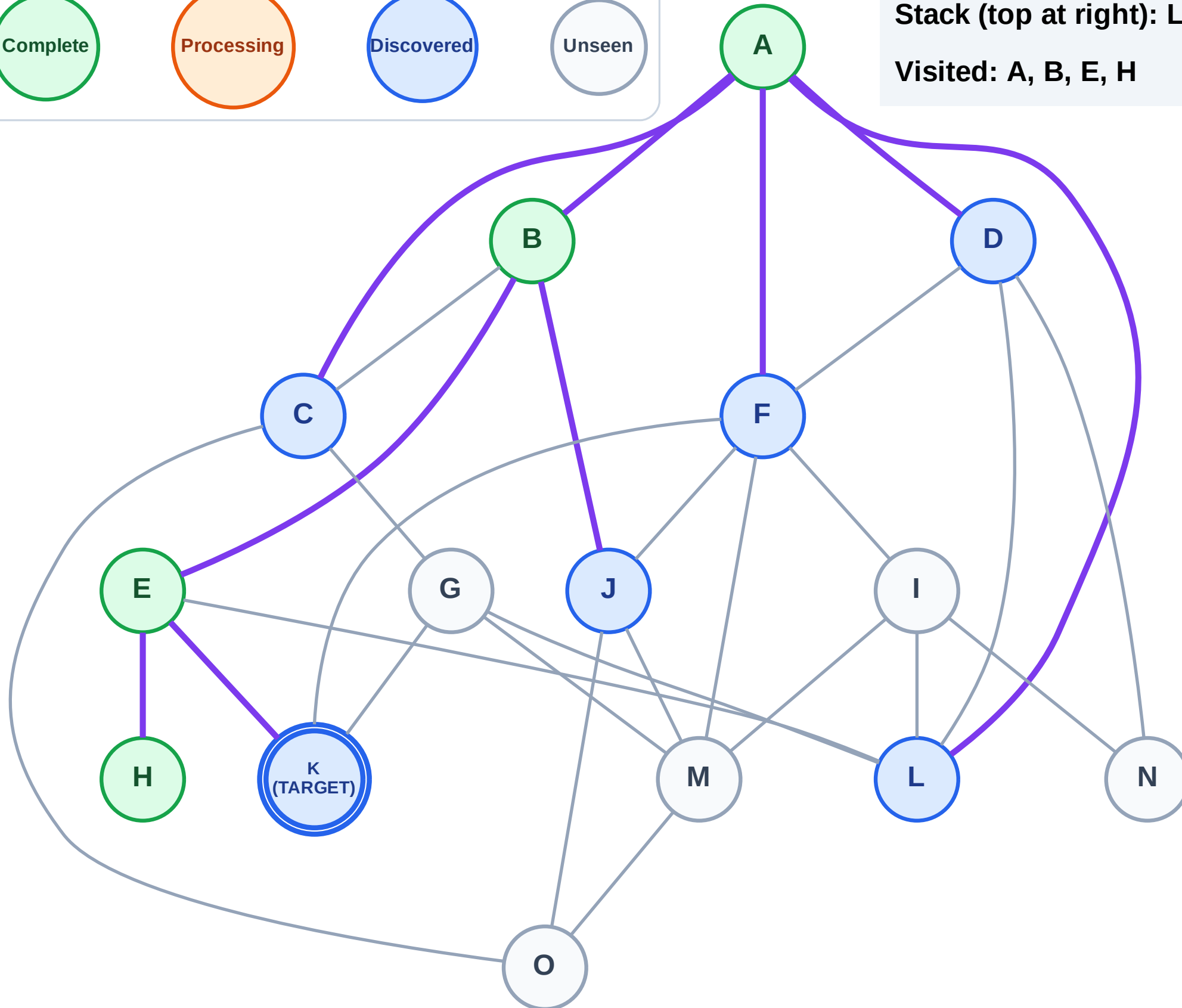
Step 9: No new neighbors from H

Legend



Stack (top at right): L | F | D | C | J | K

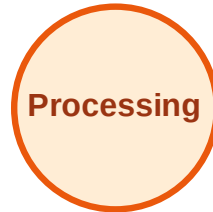
Visited: A, B, E, H



DFS Traversal

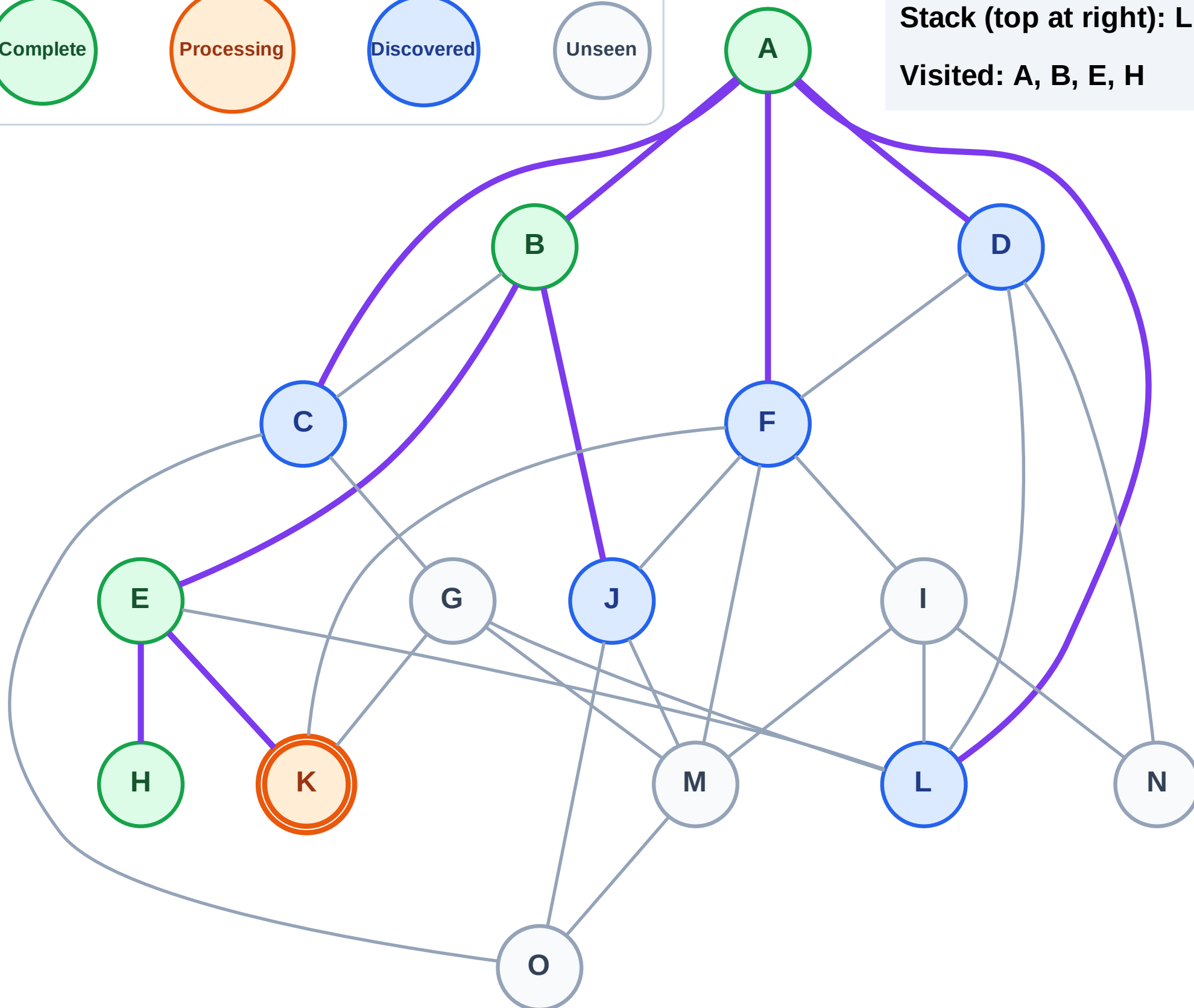
Step 10: Process node K

Legend



Stack (top at right): L | F | D | C | J

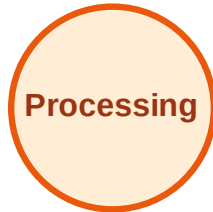
Visited: A, B, E, H



DFS Traversal

Step 11: TARGET NODE K REACHED! Path (3 hops): A → B → E → K

Legend



Stack (top at right): L | F | D | C | J

Visited: A, B, E, H, K

